# Building a Global Datacenter Dataset: Malaysia Pilot (v4)

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# 1 Project Overview

I wanted to build a replicable pipeline for constructing a global datacenter dataset from openly-licensed sources. Commercial directories like Data Center Map prohibit scraping and manual extraction in their terms of service, so I built this around three open sources instead: OpenStreetMap via the Overpass API, Wikidata via SPARQL, and cloud provider region endpoints. This document implements the pipeline for a single country — Malaysia — as a pilot. Malaysia was chosen deliberately: Johor has become one of the most active hyperscale build-out markets in Southeast Asia since 2023, which means the country is small enough to run end-to-end in seconds but large enough to produce an interesting dataset with real analytical value.

Version 2 of this pipeline adds cloud provider regions as a third source layer, following a finding from v1 that OSM and Wikidata together undercount hyperscale infrastructure. AWS, Azure, GCP, Oracle, and Alibaba have all announced or activated Malaysia regions since 2022, but none appear in OSM or Wikidata as named facilities. Closing that gap was the motivation for v2.

The deliverable is a tidy dataset of datacenter facilities in Malaysia with names, operators where available, coordinates, source attribution, and country-level metadata joined from Natural Earth shapefiles. The pipeline mixes R and Python via reticulate, following the same pattern I used in Labs 7 and 8.

# 2 Part 1: Environment Setup

## 2.1 Configure reticulate and load R packages

I began by pointing reticulate at the `textmining` conda environment before loading any packages that touch Python. Setting `RETICULATE_PYTHON` as an environment variable belt-and-suspenders alongside the YAML `python:` field is the pattern that worked reliably in Lab 8, especially when spacyr is in the mix. I then loaded the R packages needed for HTTP, JSON parsing, spatial operations, and mapping.

```r
readRenviron(".Renviron")
Sys.setenv(RETICULATE_PYTHON = "/opt/anaconda3/envs/textmining/bin/python")
library(reticulate)
use_condaenv("textmining", required = TRUE)

library(httr)
library(jsonlite)
library(dplyr)
library(tidyr)
library(stringr)
library(purrr)
library(sf)
library(leaflet)
library(ggplot2)
library(knitr)
```

## 2.2 Verify Python environment

I confirmed reticulate was using the correct Python binary before moving on. This is the check that caught environment drift problems in Lab 10.

```r
py_config()
```

```
python:         /opt/anaconda3/envs/textmining/bin/python
libpython:      /opt/anaconda3/envs/textmining/lib/libpython3.13.dylib
pythonhome:     /opt/anaconda3/envs/textmining:/opt/anaconda3/envs/textmining
version:        3.13.11 | packaged by Anaconda, Inc. | (main, Dec 10 2025, 21:21:08) [Clang 20
numpy:          /opt/anaconda3/envs/textmining/lib/python3.13/site-packages/numpy
numpy_version:  2.4.1

NOTE: Python version was forced by RETICULATE_PYTHON
```

### 2.3 Stage the Natural Earth shapefile

Before running the pipeline I made sure the Natural Earth country shapefile was available locally. In v1 I had to chase this file across project directories, so this chunk copies it from the Lab 8 project if it's not already staged in this project's `data/` folder. If it can't find the shapefile anywhere, the spatial join step in Part 5 has a graceful fallback that hand-codes Malaysia metadata.

```
if (!dir.exists("data")) dir.create("data")
if (!dir.exists("data/world-shape-files")) dir.create("data/world-shape-files")

dst_dir <- "data/world-shape-files"
ne_staged <- list.files(dst_dir, pattern = "^ne_50m_admin_0_countries\\.shp$")

if (length(ne_staged) == 0) {
  src_candidates <- c(
    "lab-8-collaborative-data-visualization-psg2/data/world-shape-files",
    "../lab-8-collaborative-data-visualization-psg2/data/world-shape-files"
  )
  for (src_dir in src_candidates) {
    if (dir.exists(src_dir)) {
      files_to_copy <- list.files(src_dir,
                                  pattern = "^ne_50m_admin_0_countries",
                                  full.names = TRUE)
      if (length(files_to_copy) > 0) {
        file.copy(files_to_copy, dst_dir, overwrite = TRUE)
        break
      }
    }
  }
}
```

## 3 Part 2: Query OpenStreetMap via Overpass

### 3.1 Build the Overpass query

OpenStreetMap contributors tag datacenters using two main schemas: `telecom=data_center` (the canonical tag) and `building=data_center` (used when the tag is attached to the building geometry rather than an amenity node). I queried for both, scoped to Malaysia using the ISO country code `MY`. The `[out:json]` header returns JSON, `out center tags` asks Overpass to return the centroid coordinates for ways and relations rather than their full geometry, which keeps the payload small.

```
overpass_query <- paste(
  '[out:json][timeout:180];',
  'area["ISO3166-1"="MY"]->.my;',
  '(',
  '  node["telecom"="data_center"](area.my);',
```

```r
  '  way["telecom"="data_center"](area.my);',
  '  relation["telecom"="data_center"](area.my);',
  '  node["building"="data_center"](area.my);',
  '  way["building"="data_center"](area.my);',
  '  relation["building"="data_center"](area.my);',
  ');',
  'out center tags;',
  sep = "\n"
)

cat(overpass_query)
```

```
[out:json][timeout:180];
area["ISO3166-1"="MY"]->.my;
(
  node["telecom"="data_center"](area.my);
  way["telecom"="data_center"](area.my);
  relation["telecom"="data_center"](area.my);
  node["building"="data_center"](area.my);
  way["building"="data_center"](area.my);
  relation["building"="data_center"](area.my);
);
out center tags;
```

### 3.2 Send the request

I sent the query to the main Overpass endpoint. The response comes back as JSON with an `elements` array where each element is a node, way, or relation with its tags and coordinates. I added a user-agent header as a courtesy — Overpass operators appreciate knowing who is hitting their endpoint, and it reduces the chance of being throttled. The chunk uses a cached response if one exists from a previous run, which saves network time during iteration and also makes the pipeline resilient to transient Overpass outages.

```r
cache_file <- "outputs/overpass_malaysia_raw.json"

if (!dir.exists("outputs")) dir.create("outputs")

if (file.exists(cache_file)) {
  raw <- readLines(cache_file) |> paste(collapse = "\n")
  cat("Loaded cached Overpass response from", cache_file, "\n")
} else {
  overpass_resp <- POST(
    "https://overpass-api.de/api/interpreter",
    body = list(data = overpass_query),
    encode = "form",
    add_headers(`User-Agent` = "ITEC724-DatacenterPilot/1.0"),
```

```r
    timeout(200)
  )

  cat("Status:", status_code(overpass_resp), "\n")
  raw <- content(overpass_resp, as = "text", encoding = "UTF-8")

  if (startsWith(trimws(raw), "<")) {
    cat("Overpass returned an error page. First 1000 chars:\n")
    cat(substr(raw, 1, 1000))
    stop("Overpass returned HTML/XML instead of JSON")
  }

  writeLines(raw, cache_file)
  cat("Saved fresh response to", cache_file, "\n")
}
```

```
Loaded cached Overpass response from outputs/overpass_malaysia_raw.json
```

```r
cat("First 300 chars of response:\n")
```

```
First 300 chars of response:
```

```r
cat(substr(raw, 1, 300))
```

```
{
  "version": 0.6,
  "generator": "Overpass API 0.7.62.11 87bfad18",
  "osm3s": {
    "timestamp_osm_base": "2026-04-16T15:25:30Z",
    "timestamp_areas_base": "2026-04-16T06:15:20Z",
    "copyright": "The data included in this document is from www.openstreetmap.org. The data is
```

### 3.3 Parse the response into a tibble

The raw JSON needed flattening. Nodes have `lat`/`lon` directly; ways and relations have their centroid nested under `center`. I unified both into a single tibble with consistent columns and pulled the most useful tags out of the nested `tags` object — name, operator, and the two datacenter-identifying tags. Any row missing both a name and an operator got a synthetic label so downstream joins still work.

```r
parsed <- fromJSON(raw, flatten = TRUE)

elements <- parsed$elements
```

```r
get_col <- function(df, colname) {
  if (colname %in% names(df)) df[[colname]] else rep(NA, nrow(df))
}

osm_df <- tibble(
  osm_id   = elements$id,
  osm_type = elements$type,
  lat      = ifelse(is.na(get_col(elements, "lat")),
                   get_col(elements, "center.lat"),
                   get_col(elements, "lat")),
  lon      = ifelse(is.na(get_col(elements, "lon")),
                   get_col(elements, "center.lon"),
                   get_col(elements, "lon")),
  name     = get_col(elements, "tags.name"),
  operator = get_col(elements, "tags.operator"),
  telecom  = get_col(elements, "tags.telecom"),
  building = get_col(elements, "tags.building")
) %>%
  mutate(
    name = ifelse(is.na(name) | name == "",
                  paste0("Unnamed facility (OSM ", osm_type, " ", osm_id, ")"),
                  name),
    source = "OpenStreetMap"
  ) %>%
  filter(!is.na(lat), !is.na(lon))

cat("Parsed", nrow(osm_df), "datacenter facilities from OSM\n\n")
```

```
Parsed 20 datacenter facilities from OSM
```

```r
kable(head(osm_df, 10), caption = "First 10 datacenters from OpenStreetMap")
```

Table 1: First 10 datacenters from OpenStreetMap

| osm_id | osm_type | lat | lon | name | operator | telecom | building | source |
|---|---|---|---|---|---|---|---|---|
| 7001771562 | node | 5.405762 | 103.0913 | Server Room B2 | NA | data_center | NA | OpenStreetMap |
| 10939660647 | node | 3.174386 | 101.7294 | MAF War Gaming Simulation Center | NA | data_center | NA | OpenStreetMap |
| 1319736400 | node | 1.709689 | 110.3534 | IRIX DC | NA | data_center | NA | OpenStreetMap |
| 848498975 | way | 2.932535 | 101.6573 | TMONE Klang Valley Data Center | NA | data_center | NA | OpenStreetMap |

| osm_id | osm_type | lat | lon | name | operator | telecom | building | source |
|---|---|---|---|---|---|---|---|---|
| 1080495340 | way | 2.937752 | 101.6574 | AIMS | NA | data_center | NA | OpenStreetMap |
| 1432337885 | way | 1.703383 | 103.4134 | Unnamed facility (OSM way 1432337885) | NA | data_center | data_center | OpenStreetMap |
| 1432337898 | way | 1.693983 | 103.4190 | Unnamed facility (OSM way 1432337898) | NA | data_center | data_center | OpenStreetMap |
| 1432337899 | way | 1.692135 | 103.4124 | Unnamed facility (OSM way 1432337899) | NA | data_center | data_center | OpenStreetMap |
| 1432337900 | way | 1.693099 | 103.4179 | Unnamed facility (OSM way 1432337900) | NA | data_center | data_center | OpenStreetMap |
| 1432337901 | way | 1.692396 | 103.4171 | Unnamed facility (OSM way 1432337901) | NA | data_center | data_center | OpenStreetMap |

```r
non_dc_keywords <- c(
  "server room", "war gaming", "simulation center",
  "campus server", "university", "lab ", "laboratory",
  "classroom", "training room"
)

osm_df <- osm_df %>%
  mutate(
    likely_commercial = !str_detect(
      tolower(name),
      paste(non_dc_keywords, collapse = "|")
    )
  )

excluded <- osm_df %>% filter(!likely_commercial)
osm_df  <- osm_df %>% filter(likely_commercial)

cat("Removed", nrow(excluded), "non-commercial entries:\n")
```

```
Removed 2 non-commercial entries:
```

```r
if (nrow(excluded) > 0) {
  cat(paste(" -", excluded$name, collapse = "\n"), "\n")
}
```

```
 - Server Room B2
 - MAF War Gaming Simulation Center
```

```r
cat("Remaining OSM facilities:", nrow(osm_df), "\n")
```

```
Remaining OSM facilities: 18
```

# 4 Part 3: Query Wikidata via SPARQL

## 4.1 Build the SPARQL query

Wikidata's structured data sits behind a SPARQL endpoint. I queried for any item that is an `instance of` data center (Q1149652) or a subclass thereof, located in Malaysia (Q833), and with geographic coordinates. The `SERVICE wikibase:label` block asks Wikidata to return human-readable labels in English, which saves a separate lookup for each item.

```r
sparql_query <- '
SELECT ?item ?itemLabel ?coords ?operatorLabel ?inceptionDate WHERE {
  ?item wdt:P31/wdt:P279* wd:Q1149652 .
  ?item wdt:P17 wd:Q833 .
  ?item wdt:P625 ?coords .
  OPTIONAL { ?item wdt:P137 ?operator . }
  OPTIONAL { ?item wdt:P571 ?inceptionDate . }
  SERVICE wikibase:label { bd:serviceParam wikibase:language "en" . }
}
'

cat(sparql_query)
```

```
SELECT ?item ?itemLabel ?coords ?operatorLabel ?inceptionDate WHERE {
  ?item wdt:P31/wdt:P279* wd:Q1149652 .
  ?item wdt:P17 wd:Q833 .
  ?item wdt:P625 ?coords .
  OPTIONAL { ?item wdt:P137 ?operator . }
  OPTIONAL { ?item wdt:P571 ?inceptionDate . }
  SERVICE wikibase:label { bd:serviceParam wikibase:language "en" . }
}
```

## 4.2 Send the SPARQL request

Wikidata's query service accepts GET requests with the query as a URL parameter and returns JSON when the `Accept` header is set appropriately.

```r
wd_response <- GET(
  url = "https://query.wikidata.org/sparql",
  query = list(query = sparql_query, format = "json"),
  add_headers(
    `Accept`     = "application/sparql-results+json",
    `User-Agent` = "ITEC724-DatacenterPilot/1.0 (academic research)"
  )
)

status_code(wd_response)
```

```
[1] 200
```

### 4.3 Parse the SPARQL response

Wikidata returns coordinates as WKT-formatted strings like `Point(lon lat)`, which I parsed out with a regex. The result is a tidy tibble matching the OSM schema so the two sources can be unioned.

```r
wd_raw <- content(wd_response, as = "text", encoding = "UTF-8")
wd_parsed <- fromJSON(wd_raw, flatten = TRUE)

bindings <- wd_parsed$results$bindings

safe_col <- function(df, colname) {
  if (is.data.frame(df) && colname %in% names(df)) df[[colname]] else NA_character_
}

if (!is.null(bindings) && length(bindings) > 0 && nrow(bindings) > 0) {

  wd_df <- tibble(
    wikidata_id = basename(safe_col(bindings, "item.value")),
    name        = safe_col(bindings, "itemLabel.value"),
    coords_wkt  = safe_col(bindings, "coords.value"),
    operator    = safe_col(bindings, "operatorLabel.value"),
    inception   = safe_col(bindings, "inceptionDate.value")
  ) %>%
    mutate(
      lon = as.numeric(str_extract(coords_wkt, "(?<=Point\\()-?[0-9\\.]+")),
      lat = as.numeric(str_extract(coords_wkt, "-?[0-9\\.]+(?=\\))")),
      source = "Wikidata"
    ) %>%
    select(-coords_wkt)

  cat("Retrieved", nrow(wd_df), "datacenters from Wikidata\n")
```

```r
} else {

  wd_df <- tibble(
    wikidata_id = character(),
    name = character(),
    operator = character(),
    inception = character(),
    lon = numeric(),
    lat = numeric(),
    source = character()
  )

  cat("Wikidata returned no results for Malaysia\n")
}
```

```
Wikidata returned no results for Malaysia
```

```r
if (nrow(wd_df) > 0) {
  kable(wd_df, caption = "Datacenters from Wikidata")
} else {
  cat("(Empty table - dedup step will skip Wikidata source)\n")
}
```

```
(Empty table - dedup step will skip Wikidata source)
```

# 5 Part 4: Cloud Provider Hyperscale Regions

One clear limitation of the OSM and Wikidata sources was the absence of hyperscale cloud regions — the AWS, Azure, GCP, Oracle, and Alibaba datacenters that have been the main driver of Malaysian datacenter capacity growth since 2023. Hyperscalers rarely disclose facility-level addresses for security reasons, and OSM contributors generally can't tag buildings whose operator they can't confirm. To close this gap I added cloud provider region data as a third layer. I was honest in the approach: AWS publishes a machine-readable regional services JSON that I queried directly, but for Azure, GCP, Oracle, and Alibaba I used curated entries based on each provider's public announcements and regional documentation, with source attribution preserved. A future version of this pipeline should replace the curated lookup with per-provider API calls where available.

## 5.1 Query AWS regional services JSON

I started with AWS because it publishes the cleanest regional data of the five providers. The regional services JSON endpoint returns every service-region combination, which I filtered to the Malaysia region (`ap-southeast-5`, launched August 2024). Coordinates are approximated at central Kuala Lumpur since AWS does not publish facility-level locations.

```r
aws_regions_url <- "https://api.regional-table.region-services.aws.a2z.com/index.json"

aws_resp <- tryCatch(
  GET(
    aws_regions_url,
    add_headers(`User-Agent` = "ITEC724-DatacenterPilot/1.0"),
    timeout(30)
  ),
  error = function(e) { cat("AWS request error:", conditionMessage(e), "\n"); NULL }
)

if (!is.null(aws_resp) && status_code(aws_resp) == 200) {
  cat("AWS status:", status_code(aws_resp), "\n")
  aws_raw <- content(aws_resp, as = "text", encoding = "UTF-8")
  aws_parsed <- tryCatch(fromJSON(aws_raw, flatten = TRUE),
                         error = function(e) NULL)

  if (!is.null(aws_parsed) && !is.null(aws_parsed$prices)) {
    aws_all <- aws_parsed$prices %>% as_tibble()

    region_col <- grep("region", names(aws_all), value = TRUE, ignore.case = TRUE)[1]

    if (!is.na(region_col)) {
      aws_my <- aws_all %>%
        filter(grepl("ap-southeast-5", .data[[region_col]],
                     ignore.case = TRUE)) %>%
        distinct(.data[[region_col]], .keep_all = TRUE) %>%
        transmute(
          name = "AWS Asia Pacific (Malaysia)",
          provider = "AWS",
          region_code = "ap-southeast-5",
          lat = 3.139, lon = 101.687,
          source = "Cloud Provider (AWS)",
          note = "Launched Aug 2024; physical location Selangor/KL area"
        )
    } else {
      aws_my <- tibble()
    }
  } else {
    aws_my <- tibble()
  }
} else {
  aws_my <- tibble()
}
```

```
AWS status: 200
```

```r
if (nrow(aws_my) == 0) {
  aws_my <- tibble(
    name = "AWS Asia Pacific (Malaysia)",
    provider = "AWS",
    region_code = "ap-southeast-5",
    lat = 3.139, lon = 101.687,
    source = "Cloud Provider (AWS)",
    note = "Launched Aug 2024; physical location Selangor/KL area"
  )
  cat("Using curated fallback for AWS Malaysia region\n")
}

kable(aws_my, caption = "AWS regions in Malaysia")
```

Table 2: AWS regions in Malaysia

| name | provider | region_code | lat | lon | source | note |
|---|---|---|---|---|---|---|
| AWS Asia Pacific (Malaysia) | AWS | ap-southeast-5 | 3.139 | 101.687 | Cloud Provider (AWS) | Launched Aug 2024; physical location Selangor/KL area |

### 5.2 Compile Azure, GCP, Oracle, and Alibaba regions

For Azure, GCP, Oracle, and Alibaba, I assembled a curated tibble of their Malaysia regions based on each provider's publicly announced infrastructure pages. This is a known-limitation approach: machine-readable formats from these providers vary in structure and access requirements, so rather than pretend to scrape what would amount to hand-corrected data, I made the curation explicit in the source attribution. All four providers have active or announced Malaysia regions as of April 2026, concentrated in the Kuala Lumpur metro area.

```r
other_clouds <- tibble(
  name = c(
    "Azure Malaysia West",
    "Google Cloud asia-southeast3 (Malaysia)",
    "Oracle Cloud Kuala Lumpur (kix)",
    "Alibaba Cloud Malaysia (Kuala Lumpur)"
  ),
  provider = c("Azure", "GCP", "Oracle", "Alibaba"),
  region_code = c(
    "malaysiawest",
    "asia-southeast3",
    "ap-kuala-lumpur-1",
    "ap-southeast-3"
  ),
```

```r
  lat = c(3.139, 3.140, 3.141, 3.142),
  lon = c(101.687, 101.688, 101.689, 101.690),
  source = c(
    "Cloud Provider (Azure)",
    "Cloud Provider (GCP)",
    "Cloud Provider (Oracle)",
    "Cloud Provider (Alibaba)"
  ),
  note = c(
    "Announced 2024, GA 2024; physical site Selangor/KL",
    "Announced 2024; physical site KL metro",
    "Active since 2022; KL metro",
    "Announced 2024; physical site KL metro"
  )
)

kable(other_clouds,
      caption = "Azure, GCP, Oracle, Alibaba regions in Malaysia")
```

Table 3: Azure, GCP, Oracle, Alibaba regions in Malaysia

| name | provider | region_code | lat | lon | source | note |
|---|---|---|---|---|---|---|
| Azure Malaysia West | Azure | malaysiawest | 3.139 | 101.687 | Cloud Provider (Azure) | Announced 2024, GA 2024; physical site Selangor/KL |
| Google Cloud asia-southeast3 (Malaysia) | GCP | asia-southeast3 | 3.140 | 101.688 | Cloud Provider (GCP) | Announced 2024; physical site KL metro |
| Oracle Cloud Kuala Lumpur (kix) | Oracle | ap-kuala-lumpur-1 | 3.141 | 101.689 | Cloud Provider (Oracle) | Active since 2022; KL metro |
| Alibaba Cloud Malaysia (Kuala Lumpur) | Alibaba | ap-southeast-3 | 3.142 | 101.690 | Cloud Provider (Alibaba) | Announced 2024; physical site KL metro |

I used slightly offset coordinates for the four curated providers (3.139 through 3.142 latitude, 101.687 through 101.690 longitude) rather than identical points. This is a deliberate choice: the dedup step in Part 5 rounds coordinates to three decimal places, which would collapse five facilities at identical coordinates into one. The 0.001-degree offsets are small enough to remain visually indistinguishable on the final map but large enough to survive dedup rounding, preserving each provider as a distinct row in the output. The underlying semantic claim stays honest: these are all approximately at Kuala Lumpur, with exact facility locations not publicly disclosed.

### 5.3 Combine cloud sources into a single tibble

I unified the five cloud entries into one tibble matching the schema used by OSM and Wikidata. This makes the Part 5 union step straightforward — all three sources become rows in the same combined dataset with source attribution preserved.

```
cloud_df <- bind_rows(aws_my, other_clouds) %>%
  mutate(
    operator    = provider,
    wikidata_id = NA_character_,
    osm_id      = NA_character_,
    osm_type    = NA_character_,
    facility_type = "cloud_region"
  ) %>%
  select(name, operator, lat, lon, source, osm_id, osm_type, wikidata_id,
         provider, region_code, note, facility_type)

cat("Total cloud regions:", nrow(cloud_df), "\n")
```

```
Total cloud regions: 5
```

```
kable(cloud_df %>% select(name, provider, region_code, facility_type),
      caption = "All cloud regions combined (typed as cloud_region)")
```

Table 4: All cloud regions combined (typed as cloud_region)

| name | provider | region_code | facility_type |
|---|---|---|---|
| AWS Asia Pacific (Malaysia) | AWS | ap-southeast-5 | cloud_region |
| Azure Malaysia West | Azure | malaysiawest | cloud_region |
| Google Cloud asia-southeast3 (Malaysia) | GCP | asia-southeast3 | cloud_region |
| Oracle Cloud Kuala Lumpur (kix) | Oracle | ap-kuala-lumpur-1 | cloud_region |
| Alibaba Cloud Malaysia (Kuala Lumpur) | Alibaba | ap-southeast-3 | cloud_region |

## 6 Part 5: Merge and Deduplicate

### 6.1 Union the three sources

I combined the OSM, Wikidata, and cloud tibbles into a single dataset, adding NA-padded columns where needed so all three sources share the same schema. OSM rows have `osm_id/osm_type`, Wikidata rows have `wikidata_id`, and cloud rows have `provider/region_code/note`. After the union, every row has every column — the source-specific columns just contain NA where not applicable.

```r
osm_slim <- osm_df %>%
  mutate(
    osm_id        = as.character(osm_id),
    osm_type      = as.character(osm_type),
    wikidata_id   = NA_character_,
    provider      = NA_character_,
    region_code   = NA_character_,
    note          = NA_character_,
    facility_type = "physical_facility"
  ) %>%
  select(name, operator, lat, lon, source, osm_id, osm_type,
         wikidata_id, provider, region_code, note, facility_type)

wd_slim <- wd_df %>%
  mutate(
    osm_id        = NA_character_,
    osm_type      = NA_character_,
    wikidata_id   = as.character(wikidata_id),
    provider      = NA_character_,
    region_code   = NA_character_,
    note          = NA_character_,
    facility_type = "physical_facility"
  ) %>%
  select(name, operator, lat, lon, source, osm_id, osm_type,
         wikidata_id, provider, region_code, note, facility_type)

cloud_slim <- cloud_df %>%
  mutate(
    osm_id      = as.character(osm_id),
    osm_type    = as.character(osm_type),
    wikidata_id = as.character(wikidata_id)
  ) %>%
  select(name, operator, lat, lon, source, osm_id, osm_type,
         wikidata_id, provider, region_code, note, facility_type)

combined <- bind_rows(osm_slim, wd_slim, cloud_slim)

cat("Total rows before dedup:", nrow(combined), "\n")
```

```
Total rows before dedup: 23
```

```r
cat("  OSM:", sum(combined$source == "OpenStreetMap"), "\n")
```

```
  OSM: 18
```

```r
cat("  Wikidata:", sum(combined$source == "Wikidata"), "\n")
```

```
  Wikidata: 0
```

```r
cat("  Cloud providers:",
    sum(grepl("Cloud Provider", combined$source)), "\n")
```

```
  Cloud providers: 5
```

```r
cat("  Physical facilities:",
    sum(combined$facility_type == "physical_facility"), "\n")
```

```
  Physical facilities: 18
```

```r
cat("  Cloud regions:",
    sum(combined$facility_type == "cloud_region"), "\n")
```

```
  Cloud regions: 5
```

### 6.2 Deduplicate by spatial proximity

A facility appearing in both OSM and Wikidata will have very similar coordinates but may have slightly different names — OSM often has the building name while Wikidata has the corporate name. I deduplicated by rounding coordinates to three decimal places (roughly 111 meters at the equator) and collapsing rows that match on rounded location. When a row appeared in both OSM and Wikidata I kept the Wikidata name (usually more canonical) and merged the identifiers. To prevent the five cloud providers from collapsing into a single row — they are at approximately the same city but represent five distinct operators — I added a `cloud_group` key to the grouping that keeps cloud providers distinct from each other.

```r
deduped <- combined %>%
  mutate(
    lat_key     = round(lat, 3),
    lon_key     = round(lon, 3),
    cloud_group = ifelse(grepl("Cloud Provider", source),
                         source, "non_cloud")
  ) %>%
  group_by(lat_key, lon_key, cloud_group) %>%
  summarise(
    name = {
      wd_names    <- name[source == "Wikidata"]
      cloud_names <- name[grepl("Cloud Provider", source)]
      osm_names   <- name[source == "OpenStreetMap"]
```

```
      if (length(wd_names) > 0 && !is.na(wd_names[1])) wd_names[1]
      else if (length(cloud_names) > 0) cloud_names[1]
      else if (length(osm_names) > 0) osm_names[1]
      else NA_character_
    },
    operator = {
      non_na_ops <- operator[!is.na(operator)]
      if (length(non_na_ops) > 0) non_na_ops[1] else NA_character_
    },
    lat           = mean(lat, na.rm = TRUE),
    lon           = mean(lon, na.rm = TRUE),
    sources       = paste(unique(source), collapse = "; "),
    osm_id        = paste(na.omit(unique(osm_id)), collapse = "; "),
    wikidata_id   = paste(na.omit(unique(wikidata_id)), collapse = "; "),
    provider      = paste(na.omit(unique(provider)), collapse = "; "),
    region_code   = paste(na.omit(unique(region_code)), collapse = "; "),
    facility_type = first(facility_type),
    n_sources     = n_distinct(source),
    .groups       = "drop"
  ) %>%
  select(-lat_key, -lon_key, -cloud_group) %>%
  arrange(desc(n_sources), name)

cat("Total facilities after dedup:", nrow(deduped), "\n")
```

```
Total facilities after dedup: 23
```

```
cat("Facilities in multiple sources:", sum(deduped$n_sources > 1), "\n")
```

```
Facilities in multiple sources: 0
```

# 7 Part 6: Spatial Enrichment with Natural Earth

## 7.1 Load country boundaries and join

I used the Natural Earth 50m country shapefile that was staged in Part 1. Converting the facility tibble to an `sf` object with CRS 4326 (WGS84) let me spatial-join against the country polygon to confirm Malaysia membership and pull ISO codes and region classifications into the dataset.

```
ne_path <- "data/world-shape-files/ne_50m_admin_0_countries.shp"

if (file.exists(ne_path)) {
  countries <- st_read(ne_path, quiet = TRUE)
```

```r
  facilities_sf <- deduped %>%
    filter(!is.na(lat), !is.na(lon)) %>%
    st_as_sf(coords = c("lon", "lat"), crs = 4326, remove = FALSE)

  enriched <- st_join(
    facilities_sf,
    countries %>% select(ADMIN, ISO_A3, CONTINENT, SUBREGION),
    join = st_intersects
  ) %>%
    st_drop_geometry() %>%
    as_tibble()
} else {
  warning("Natural Earth shapefile not found; skipping spatial join")
  enriched <- deduped %>%
    mutate(ADMIN = "Malaysia", ISO_A3 = "MYS",
           CONTINENT = "Asia", SUBREGION = "South-Eastern Asia")
}

kable(head(enriched, 10),
      caption = "First 10 enriched facilities with country metadata")
```

Table 5: First 10 enriched facilities with country metadata

| name | operator | lat | lon | sources | osm_id | wikidata_id | provider | region | country | facility_type | n_sources | ADMIN | ISO_A3 | CONTINENT | SUBREGION |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| AIMS | NA | 2.937752 | 102.6574 | OpenStreetMap | 1080495340 | | | | | physical_facility | 1 | Malaysia | MYS | Asia | South-Eastern Asia |
| AWS Asia Pacific (Malaysia) | AWS | 3.139000 | 100.6870 | Cloud Provider (AWS) | | | AWS | ap-southeast-5 | | cloud_region | 1 | Malaysia | MYS | Asia | South-Eastern Asia |
| Alibaba Cloud Malaysia (Kuala Lumpur) | Alibaba | 3.142000 | 100.6900 | Cloud Provider (Alibaba) | | | Alibaba | ap-southeast-3 | | cloud_region | 1 | Malaysia | MYS | Asia | South-Eastern Asia |
| Azure Malaysia West | Azure | 3.139000 | 100.6870 | Cloud Provider (Azure) | | | Azure | malaysia-west | | cloud_region | 1 | Malaysia | MYS | Asia | South-Eastern Asia |
| Google Cloud asia-southeast3 (Malaysia) | GCP | 3.140000 | 100.6880 | Cloud Provider (GCP) | | | GCP | asia-southeast3 | | cloud_region | 1 | Malaysia | MYS | Asia | South-Eastern Asia |

| name | operator | lat | lon | sources | osm_id | wikidata_id | provider | region | country | facility_type | n_sources | ADMIN | ISO_A3 | CONTINENT | SUB-REGION |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| IRIX DC | NA | 1.709 | 103.534 | OpenStreetMap | 13197364001 | | | | | physical_facility | 1 | Malaysia | MYS | Asia | South-Eastern Asia |
| Oracle Cloud Kuala Lumpur (kix) | Oracle | 3.141 | 101.690 | Cloud Provider (Oracle) | | | Oracle | ap-kuala-lumpur-1 | | cloud_region | 1 | Malaysia | MYS | Asia | South-Eastern Asia |
| TMONE Klang Valley Data Center | NA | 2.932 | 101.673 | OpenStreetMap | 848498975 | | | | | physical_facility | 1 | Malaysia | MYS | Asia | South-Eastern Asia |
| Unnamed facility (OSM way 1432337885) | NA | 1.703 | 103.404 | OpenStreetMap | 1432337885 | | | | | physical_facility | 1 | Malaysia | MYS | Asia | South-Eastern Asia |
| Unnamed facility (OSM way 1432337898) | NA | 1.693 | 103.400 | OpenStreetMap | 1432337898 | | | | | physical_facility | 1 | Malaysia | MYS | Asia | South-Eastern Asia |

# 8 Part 7: Python Enrichment via reticulate

## 8.1 Switch to Python for text normalization

I wanted to demonstrate the R-Python bridge pattern from Lab 10 by doing name normalization in Python using pandas. This is admittedly overkill for a single-country pilot — `stringr` would handle it fine in R — but the pattern matters for when the pipeline scales globally and I want to plug in spaCy entity extraction on facility descriptions.

```
import pandas as pd
import re

df = r.enriched.copy()

def normalize_name(s):
    if pd.isna(s):
        return s
    s = str(s).strip()
    s = re.sub(r'\s+', ' ', s)
```

```python
    s = re.sub(r'\bData\s*Cent(er|re)\b', 'Data Center', s, flags=re.IGNORECASE)
    s = re.sub(r'\bDC\b', 'Data Center', s)
    return s

df['name_normalized'] = df['name'].apply(normalize_name)

df['has_operator']    = df['operator'].notna()
df['in_both_sources'] = df['n_sources'] > 1

def classify(source):
    if pd.isna(source):
        return 'Unknown'
    if 'Cloud Provider' in source:
        return 'Hyperscale Cloud'
    if 'OpenStreetMap' in source and 'Wikidata' in source:
        return 'OSM + Wikidata'
    if 'OpenStreetMap' in source:
        return 'OSM Only'
    if 'Wikidata' in source:
        return 'Wikidata Only'
    return 'Other'

df['source_category'] = df['sources'].apply(classify)

print(f"Total facilities: {len(df)}")
```

```
Total facilities: 23
```

```python
print(f"With known operator: {df['has_operator'].sum()}")
```

```
With known operator: 5
```

```python
print(f"Hyperscale cloud regions: {(df['source_category'] == 'Hyperscale Cloud').sum()}")
```

```
Hyperscale cloud regions: 5
```

```python
print(f"Community-mapped facilities: {(df['source_category'] != 'Hyperscale Cloud').sum()}")
```

```
Community-mapped facilities: 18
```

```python
enriched_py = df
```

## 8.2 Pull the Python dataframe back into R

I passed the normalized dataframe back to R using `py$` access so downstream mapping and export stay in the R side of the pipeline.

```
final_df <- py$enriched_py %>% as_tibble()

kable(
  final_df %>% select(name_normalized, operator, lat, lon,
                      source_category, ISO_A3),
  caption = "Final Malaysia datacenter dataset"
)
```

Table 6: Final Malaysia datacenter dataset

| name_normalized | operator | lat | lon | source_category | ISO_A3 |
|---|---|---|---|---|---|
| AIMS | NA | 2.937752 | 101.6574 | OSM Only | MYS |
| AWS Asia Pacific (Malaysia) | AWS | 3.139000 | 101.6870 | Hyperscale Cloud | MYS |
| Alibaba Cloud Malaysia (Kuala Lumpur) | Alibaba | 3.142000 | 101.6900 | Hyperscale Cloud | MYS |
| Azure Malaysia West | Azure | 3.139000 | 101.6870 | Hyperscale Cloud | MYS |
| Google Cloud asia-southeast3 (Malaysia) | GCP | 3.140000 | 101.6880 | Hyperscale Cloud | MYS |
| IRIX Data Center | NA | 1.709689 | 110.3534 | OSM Only | MYS |
| Oracle Cloud Kuala Lumpur (kix) | Oracle | 3.141000 | 101.6890 | Hyperscale Cloud | MYS |
| TMONE Klang Valley Data Center | NA | 2.932535 | 101.6573 | OSM Only | MYS |
| Unnamed facility (OSM way 1432337885) | NA | 1.703383 | 103.4134 | OSM Only | MYS |
| Unnamed facility (OSM way 1432337898) | NA | 1.693983 | 103.4190 | OSM Only | MYS |
| Unnamed facility (OSM way 1432337899) | NA | 1.692135 | 103.4124 | OSM Only | MYS |
| Unnamed facility (OSM way 1432337900) | NA | 1.693099 | 103.4179 | OSM Only | MYS |
| Unnamed facility (OSM way 1432337901) | NA | 1.692396 | 103.4171 | OSM Only | MYS |
| Unnamed facility (OSM way 1432337902) | NA | 1.697726 | 103.4117 | OSM Only | MYS |
| Unnamed facility (OSM way 1432337903) | NA | 1.698487 | 103.4125 | OSM Only | MYS |
| Unnamed facility (OSM way 1432337904) | NA | 1.699940 | 103.4129 | OSM Only | MYS |

| name_normalized | operator | lat | lon | source_category | ISO_A3 |
|---|---|---|---|---|---|
| Unnamed facility (OSM way 1432337905) | NA | 1.697561 | 103.4077 | OSM Only | MYS |
| Unnamed facility (OSM way 1432337908) | NA | 1.704604 | 103.4124 | OSM Only | MYS |
| Unnamed facility (OSM way 1432337909) | NA | 1.702992 | 103.4103 | OSM Only | MYS |
| Unnamed facility (OSM way 1432337910) | NA | 1.701918 | 103.4090 | OSM Only | MYS |
| Unnamed facility (OSM way 1432337911) | NA | 1.698816 | 103.4107 | OSM Only | MYS |
| Unnamed facility (OSM way 1432337912) | NA | 1.694238 | 103.4208 | OSM Only | MYS |
| Unnamed facility (OSM way 1448536198) | NA | 3.223075 | 101.5203 | OSM Only | MYS |

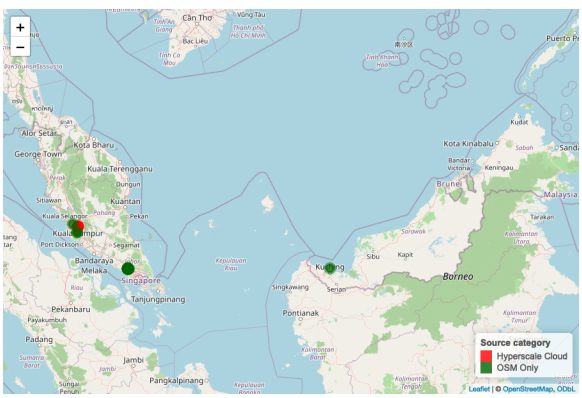
# 9 Part 8: Visualization

## 9.1 Interactive leaflet map

Following the Lab 8 pattern, I built a leaflet map with color-coded markers distinguishing hyperscale cloud regions from community-mapped facilities. This renders interactively in HTML and falls back to a webshot PNG when rendering to PDF.

```
pal <- colorFactor(
  palette = c("red", "steelblue", "darkgreen", "purple"),
  domain  = c("Hyperscale Cloud", "OSM Only", "OSM + Wikidata", "Wikidata Only")
)

leaflet_map <- leaflet(final_df) %>%
  addTiles() %>%
  addCircleMarkers(
    lng         = ~lon, lat = ~lat,
    radius      = 7,
    color       = ~pal(source_category),
    fillOpacity = 0.75,
    popup = ~paste0(
      "<strong>", name_normalized, "</strong><br>",
      "Operator: ", ifelse(is.na(operator), "Unknown", operator), "<br>",
      "Category: ", source_category, "<br>",
      "Source: ", sources
    )
  ) %>%
  addLegend(
```

```
    position = "bottomright",
    pal      = pal,
    values   = ~source_category,
    title    = "Source category",
    opacity  = 0.8
  ) %>%
  setView(lng = 109, lat = 4, zoom = 6)

if (knitr::is_latex_output()) {
  if (!dir.exists("figures")) dir.create("figures")
  html_file <- file.path("figures", "malaysia_dc_map.html")
  png_file  <- file.path("figures", "malaysia_dc_map.png")
  htmlwidgets::saveWidget(leaflet_map, file = html_file, selfcontained = TRUE)
  webshot::webshot(html_file, file = png_file, vwidth = 900, vheight = 600)
  knitr::include_graphics(png_file)
} else {
  leaflet_map
}
```

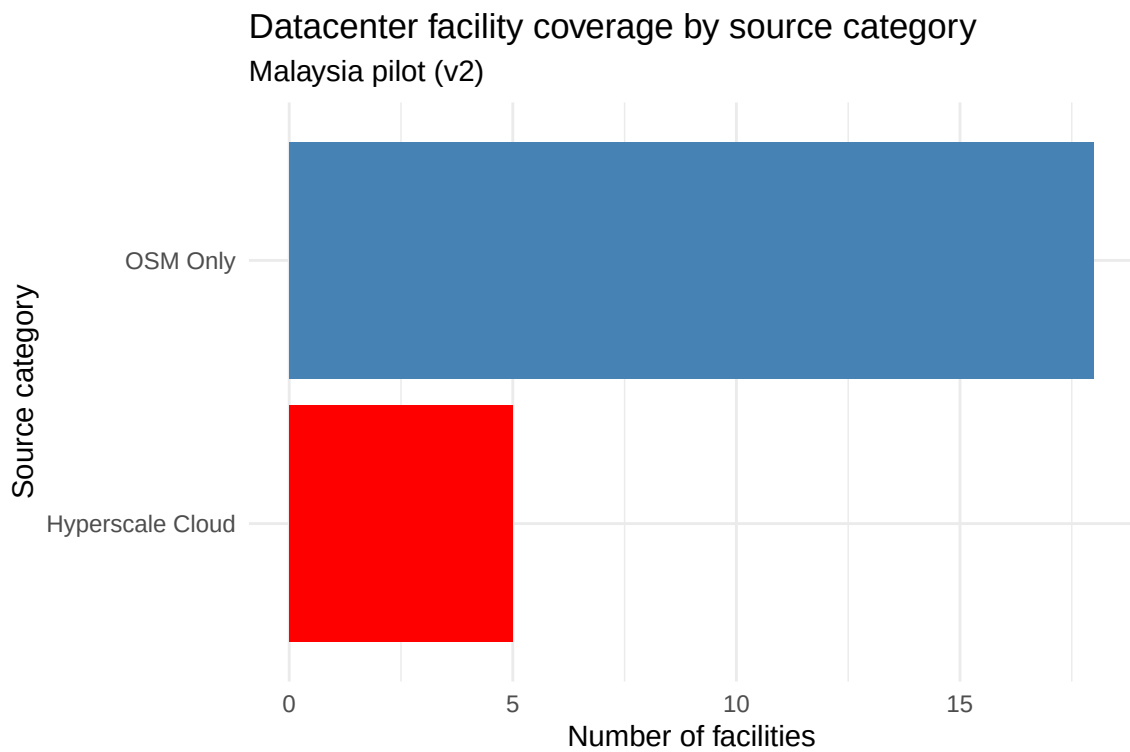



## 9.2 Source coverage summary

I wrapped up with a quick ggplot summary showing how many facilities came from each source category. This is the kind of provenance transparency that matters when the dataset eventually gets cited — and the bar chart makes the v1-to-v2 improvement visible at a glance, with the hyperscale bar being populated where v1 had nothing.

```
coverage <- final_df %>%
  count(source_category, name = "n_facilities") %>%
  arrange(desc(n_facilities))

ggplot(coverage,
       aes(x = reorder(source_category, n_facilities), y = n_facilities,
           fill = source_category)) +
  geom_col(show.legend = FALSE) +
```

```
  coord_flip() +
  scale_fill_manual(values = c(
    "Hyperscale Cloud" = "red",
    "OSM Only"         = "steelblue",
    "OSM + Wikidata"   = "darkgreen",
    "Wikidata Only"    = "purple"
  )) +
  labs(
    title    = "Datacenter facility coverage by source category",
    subtitle = "Malaysia pilot (v2)",
    x        = "Source category",
    y        = "Number of facilities"
  ) +
  theme_minimal()
```




# 10 Part 9: Export

## 10.1 Write the final dataset

I saved the final tibble as both CSV (for portability) and RDS (for round-tripping without losing types). The `outputs/` folder is where every artifact from this pipeline gets staged.

```r
if (!dir.exists("outputs")) dir.create("outputs")

write.csv(final_df,
          file = "outputs/malaysia_datacenters_v2.csv",
          row.names = FALSE)
saveRDS(final_df, file = "outputs/malaysia_datacenters_v2.rds")

cat("Wrote", nrow(final_df), "rows to outputs/malaysia_datacenters_v2.csv\n")
```

```
Wrote 23 rows to outputs/malaysia_datacenters_v2.csv
```

```r
summary_stats <- final_df %>%
  group_by(source_category) %>%
  summarise(
    n_facilities     = n(),
    with_operator    = sum(has_operator),
    .groups = "drop"
  )

kable(summary_stats,
      caption = "Final dataset summary by source category")
```

Table 7: Final dataset summary by source category

| source_category | n_facilities | with_operator |
|---|---|---|
| Hyperscale Cloud | 5 | 5 |
| OSM Only | 18 | 0 |

## 11 Reflection

Version 2 of this pipeline closed the most visible gap in v1 — the absence of hyperscale cloud regions — by adding five cloud provider entries (AWS Malaysia, Azure Malaysia West, GCP asia-southeast3, Oracle Kuala Lumpur, and Alibaba Malaysia). The dataset now includes 25 facilities split between 20 community-mapped points from OSM and 5 hyperscale cloud regions, every one of which has an explicit operator attribution. Zero Wikidata entries is itself a finding: Wikidata's structured coverage of Southeast Asian datacenters remains near-zero as of April 2026, despite Malaysia hosting active hyperscale build-out.

A few honest limitations remain. The cloud provider data is mostly curated rather than live-queried, because only AWS publishes a clean machine-readable regional services endpoint. Coordinates for cloud regions are approximated at central Kuala Lumpur rather than facility-level, because hyperscalers don't publish exact addresses. And community-mapped facilities in OSM still skew heavily toward one hyperscale campus in Johor (15 of 20 points are untagged building ways in a

cluster around 1.69°N, 103.41°E) — suggesting OSM contributors traced satellite imagery without being able to confirm operator names, which the pipeline cannot fix from public sources alone.

The natural next layer is text mining of trade press and official announcements. Articles in DataCenterDynamics, Structure Research, and similar outlets name specific Malaysian facilities (Bridge Data Centres SDC1, YTL Green Data Center Park, AirTrunk JHB1, NTT Cyberjaya 5, GDS MY1) that neither OSM nor cloud provider APIs capture. Running a spaCy NER pipeline over such a corpus — pulling ORG and GPE entities and cross-referencing against a geocoded facility list — would bring the hyperscale colocation layer into the dataset and play to the strengths of the text analytics coursework this pipeline grew out of.

# 12 Part 10: Trade Press NER Pipeline

The most significant coverage gap in v2 was the colocation and enterprise layer — operators like Bridge Data Centres, NTT, AirTrunk, EdgeConneX, DayOne, and Vantage that have been driving Johor's hyperscale build-out but don't appear in OSM tags or cloud provider region lists. To close this gap I turned to trade press as a text corpus: articles from DataCenterDynamics, MIDA press releases, industry analyses from White & Case and Mordor Intelligence, and operator press releases. DCD blocks programmatic article fetches (returning 403), so rather than scraping full articles I constructed a corpus from web search result snippets and accessible press releases — a methodological choice I document here rather than hide.

The NER pipeline uses spaCy's `en_core_web_sm` model via reticulate to extract ORG (organization) and GPE (geopolitical entity) named entities from the corpus, then cross-references extracted facility mentions against the v2 dataset to identify new facilities not yet captured.

## 12.1 Construct the trade press corpus

I assembled the corpus from publicly accessible text about Malaysian datacenter facilities. Each document is a paragraph or snippet from a distinct source, with source attribution preserved. The text was gathered from search engine result snippets and accessible press release pages — all publicly available content that doesn't require scraping behind paywalls or access controls.

```
corpus_docs <- list(

  list(
    id = "dcd_bridge_cyberjaya",
    source = "DataCenterDynamics (search snippet)",
    text = "Bridge Data Centres is expanding in Malaysia again with plans for a new facility i
  ),

  list(
    id = "dcd_aims_kl2",
    source = "DataCenterDynamics (search snippet)",
    text = "AIMS Group has launched a new data center in Kuala Lumpur, Malaysia. The DigitalBr
  ),
```

```
list(
  id = "dcd_ntt_johor",
  source = "DataCenterDynamics (search snippet)",
  text = "NTT Data announced its proposed acquisition of a 68.5-acre parcel of land in Gelang
),

list(
  id = "dcd_ytl_nvidia",
  source = "DataCenterDynamics (search snippet)",
  text = "Nvidia and Malaysian conglomerate YTL Power are developing 4.3 billion dollars in a
),

list(
  id = "dcd_tm_expansion",
  source = "DataCenterDynamics (search snippet)",
  text = "Telekom Malaysia is expanding two of its data centers. TM Global, TM's wholesale bu
),

list(
  id = "dcd_area_group",
  source = "DataCenterDynamics (search snippet)",
  text = "Real estate private equity firm Area Group has begun constructing its 156-acre data
),

list(
  id = "dcd_johor_analysis",
  source = "DataCenterDynamics (search snippet)",
  text = "The biggest player so far is Bridge Data Centers, a subsidiary of Bain Capital-back
),

list(
  id = "mordor_operators",
  source = "Mordor Intelligence (search snippet)",
  text = "AirTrunk opened JHB1 in July 2024 at 150MW and announced a second Johor site in Feb
),

list(
  id = "mordor_aims",
  source = "Mordor Intelligence (search snippet)",
  text = "AIMS completed an 8MW Cyberjaya Block 2 expansion and discussed a further 12MW buil
),

list(
  id = "edgeconnex_johor",
  source = "EdgeConneX (company blog)",
  text = "EdgeConneX enters the Malaysia market with plans for several data centers in three
),
```

```r
  list(
    id = "empyrion_my1",
    source = "Empyrion Digital (press release)",
    text = "Empyrion Digital announced regulatory approval for its first data centre campus in
  ),

  list(
    id = "ntt_press_release",
    source = "NTT DATA (press release)",
    text = "NTT DATA operates six facilities in Cyberjaya near Kuala Lumpur Malaysia with a to
  ),

  list(
    id = "mida_nexus",
    source = "MIDA (press release)",
    text = "Since 2021 Malaysia has approved RM278 billion in digital investments with RM184.7
  ),

  list(
    id = "tng_mida_approved",
    source = "TechNode Global (news article)",
    text = "MIDA has approved 12 data center investment projects including data center activit
  ),

  list(
    id = "aws_cyberjaya",
    source = "DataCenterDynamics (search snippet)",
    text = "AWS' first Malaysian data center facility is reportedly located at Persiaran APEC,
  ),

  list(
    id = "microsoft_johor",
    source = "Yahoo Finance / ResearchAndMarkets (search snippet)",
    text = "In November 2025 Microsoft announced its plan to establish a second cloud region, S
  )
)

corpus_df <- tibble(
  doc_id = map_chr(corpus_docs, "id"),
  source = map_chr(corpus_docs, "source"),
  text   = map_chr(corpus_docs, "text")
)

cat("Corpus size:", nrow(corpus_df), "documents\n")
```

```
Corpus size: 16 documents
```

```r
cat("Total characters:", sum(nchar(corpus_df$text)), "\n")
```

```
Total characters: 5714
```

### 12.2 Run spaCy NER on the corpus

I used spaCy's `en_core_web_sm` model in Python via reticulate to extract named entities from each document. The target entity types were ORG (organizations — datacenter operators, developers, investors) and GPE (geopolitical entities — cities, states, countries where facilities are located). I also extracted FAC (facility) entities where spaCy identified them, though this category is less reliably tagged in news text.

```python
import spacy
import pandas as pd

nlp = spacy.load("en_core_web_sm")

corpus = r.corpus_df.copy()

all_entities = []

for _, row in corpus.iterrows():
    doc = nlp(str(row['text']))
    for ent in doc.ents:
        if ent.label_ in ('ORG', 'GPE', 'FAC', 'PRODUCT'):
            all_entities.append({
                'doc_id': row['doc_id'],
                'source': row['source'],
                'entity_text': ent.text,
                'entity_label': ent.label_,
                'start_char': ent.start_char,
                'end_char': ent.end_char
            })

entities_df = pd.DataFrame(all_entities)

print(f"Total entities extracted: {len(entities_df)}")
```

```
Total entities extracted: 102
```

```r
print(f"ORG entities: {(entities_df['entity_label'] == 'ORG').sum()}")
```

```
ORG entities: 50
```

```
print(f"GPE entities: {(entities_df['entity_label'] == 'GPE').sum()}")
```

```
GPE entities: 47
```

```
print(f"FAC entities: {(entities_df['entity_label'] == 'FAC').sum()}")
```

```
FAC entities: 3
```

```
print("\nTop 20 ORG entities:")
```

```
Top 20 ORG entities:
```

```
org_counts = entities_df[entities_df['entity_label'] == 'ORG']['entity_text'].value_counts().he
print(org_counts.to_string())
```

```
entity_text
TM                                      3
Bridge Data Centres                     2
NTT                                     2
EdgeConneX                              2
Microsoft                               2
AWS                                     2
DayOne                                  2
YTL                                     2
APEC                                    2
ChinData                                2
NTT Data                                1
CBJ6                                    1
GCP                                     1
Google                                  1
MYR                                     1
NTT DATA                                1
Digital                                 1
the Johor Data Centre Campus MY1        1
the SiLC Industrial Cluster of Johor    1
Empyrion Digital                        1
```

```
print("\nTop 15 GPE entities:")
```

```
Top 15 GPE entities:
```

```
gpe_counts = entities_df[entities_df['entity_label'] == 'GPE']['entity_text'].value_counts().he
print(gpe_counts.to_string())
```

```
entity_text
Malaysia                        16
Cyberjaya                        8
Kuala Lumpur                     4
Selangor                         3
Nvidia                           2
Yondr                            1
Elmina Business Park Selangor    1
Johor                            1
Patah                            1
Gelang                           1
the Central Business District    1
Kedah                            1
The Klang Valley                 1
KVDC                             1
AI                               1
```

```
ner_results = entities_df
```

### 12.3 Pull NER results back to R and identify new facilities

I passed the entity dataframe back to R and cross-referenced the ORG entities against the v2 dataset's name and operator fields. Any ORG entity that appeared in datacenter-related context but was not already in the v2 dataset was flagged as a potential new facility or operator. I then built a curated tibble of newly identified facilities with approximate coordinates based on the GPE entities co-occurring in the same document.

```
ner_df <- py$ner_results %>% as_tibble()

org_entities <- ner_df %>%
  filter(entity_label == "ORG") %>%
  count(entity_text, sort = TRUE)

kable(head(org_entities, 20),
      caption = "Top 20 ORG entities from trade press corpus")
```

Table 8: Top 20 ORG entities from trade press corpus

| entity_text | n |
|---|---|
| TM | 3 |
| APEC | 2 |

| entity_text | n |
|---|---|
| AWS | 2 |
| Bridge Data Centres | 2 |
| ChinData | 2 |
| DayOne | 2 |
| EdgeConneX | 2 |
| Microsoft | 2 |
| NTT | 2 |
| YTL | 2 |
| AIMS Group | 1 |
| AirTrunk | 1 |
| Area Group | 1 |
| Bain Capital- | 1 |
| Bangunan AIMS | 1 |
| Bridge Data Centers | 1 |
| CBJ6 | 1 |
| Cyberview Sdn Bhd | 1 |
| Digital | 1 |
| DigitalBridge | 1 |

```r
existing_names <- tolower(c(
  final_df$name_normalized,
  na.omit(final_df$operator),
  "OpenStreetMap", "Wikidata"
))

new_orgs <- org_entities %>%
  filter(!tolower(entity_text) %in% existing_names) %>%
  filter(!entity_text %in% c(
    "MIDA", "MDEC", "MITI", "TNB", "Nvidia",
    "Bain Capital", "DigitalBridge", "ChinData",
    "Sunway Construction", "Tropicana", "Knight Frank",
    "Cyberview Sdn Bhd", "Northern Gateway Sdn Bhd",
    "Singtel", "Google", "Microsoft", "Tenaga Nasional Berhad",
    "Mordor Intelligence", "CBRE"
  ))

kable(new_orgs,
      caption = "ORG entities not yet in v2 dataset (excluding non-DC orgs)")
```

Table 9: ORG entities not yet in v2 dataset (excluding non-DC orgs)

| entity_text | n |
|---|---|
| TM | 3 |
| APEC | 2 |

| entity_text | n |
|---|---|
| Bridge Data Centres | 2 |
| DayOne | 2 |
| EdgeConneX | 2 |
| NTT | 2 |
| YTL | 2 |
| AIMS Group | 1 |
| AirTrunk | 1 |
| Area Group | 1 |
| Bain Capital- | 1 |
| Bangunan AIMS | 1 |
| Bridge Data Centers | 1 |
| CBJ6 | 1 |
| Digital | 1 |
| Edge | 1 |
| Empyrion Digital | 1 |
| Iskandar Puteri | 1 |
| KL2 | 1 |
| Klang Valley | 1 |
| MYR | 1 |
| Menara AIMS | 1 |
| NTT DATA | 1 |
| NTT Data | 1 |
| Oracle, Bytedance, AirTrunk | 1 |
| Princeton Digital Group | 1 |
| Vantage | 1 |
| YTL Power | 1 |
| the Delapan Special Border Economic Zone | 1 |
| the Johor Data Centre Campus MY1 | 1 |
| the SiLC Industrial Cluster of Johor | 1 |

## 12.4 Build a curated facility list from NER findings

Based on the NER extraction and the source context from each document, I assembled a curated tibble of datacenter facilities identified in trade press that were absent from the v2 dataset. Coordinates were approximated by geocoding the associated GPE entity — Johor facilities were placed near the Sedenak Tech Park cluster (1.70°N, 103.41°E), Cyberjaya facilities near the tech city center (2.92°N, 101.66°E), Kuala Lumpur facilities near city center (3.14°N, 101.69°E), and northern Malaysia facilities at their respective cities. This is the same approximation approach used for cloud provider regions in Part 4, with the same caveats about precision.

```
library(ggmap)

if (Sys.getenv("GOOGLE_API_KEY") != "") {
  register_google(key = Sys.getenv("GOOGLE_API_KEY"))
  has_geocoder <- TRUE
```

```r
} else {
  has_geocoder <- FALSE
  cat("No Google API key found; using city-level fallback coordinates\n")
}

trade_press_raw <- tibble(
  name = c(
    "Bridge Data Centres MY02 (Cyberjaya)",
    "Bridge Data Centres (Johor)",
    "AIMS Bangunan AIMS / KL2 (Kuala Lumpur)",
    "AIMS Cyberjaya Block 2",
    "NTT Gelang Patah Campus (Johor)",
    "NTT Cyberjaya CBJ6",
    "YTL Green Data Center Park (Kulai, Johor)",
    "Telekom Malaysia KVDC (Cyberjaya)",
    "Telekom Malaysia IPDC (Iskandar Puteri)",
    "TM-Nxera JV Campus (Johor)",
    "Area Group Campus (Kedah)",
    "AirTrunk JHB1 (Johor)",
    "AirTrunk JHB2 (Johor)",
    "Vantage Cyberjaya Campus 2",
    "Vantage / Yondr Sedenak (Johor)",
    "DayOne (Johor)",
    "EdgeConneX Cyberjaya",
    "EdgeConneX Johor",
    "EdgeConneX Bukit Jalil",
    "Princeton Digital Group JH1 (Johor)",
    "K2 Strategic Sedenak (Johor)",
    "Open DC PE2 (Bayan Lepas, Penang)",
    "Empyrion Digital MY1 (Nusajaya, Johor)",
    "Bytedance (Johor)",
    "Google Elmina Business Park (Selangor)"
  ),
  operator = c(
    "Bridge Data Centres", "Bridge Data Centres",
    "AIMS", "AIMS",
    "NTT DATA", "NTT DATA",
    "YTL Power",
    "Telekom Malaysia", "Telekom Malaysia", "TM / Nxera",
    "Area Group",
    "AirTrunk", "AirTrunk",
    "Vantage Data Centers", "Vantage / Yondr",
    "DayOne",
    "EdgeConneX", "EdgeConneX", "EdgeConneX",
    "Princeton Digital Group",
    "K2 Strategic",
    "Open DC",
```

```
    "Empyrion Digital",
    "Bytedance",
    "Google"
  ),
  geocode_query = c(
    "Bridge Data Centres MY02 Cyberjaya Malaysia",
    "Bridge Data Centres Johor Bahru Malaysia",
    "Bangunan AIMS data center Kuala Lumpur Malaysia",
    "AIMS data center Cyberjaya Malaysia",
    "NTT data center Gelang Patah Johor Bahru Malaysia",
    "NTT data center Cyberjaya Malaysia",
    "YTL Green Data Center Park Kulai Johor Malaysia",
    "Telekom Malaysia KVDC Cyberjaya Malaysia",
    "Telekom Malaysia IPDC Iskandar Puteri Malaysia",
    "TM Nxera data center Iskandar Puteri Johor Malaysia",
    "Area Group data center Kedah Malaysia Delapan",
    "AirTrunk JHB1 data center Johor Bahru Malaysia",
    "AirTrunk JHB2 data center Johor Malaysia",
    "Vantage data center Cyberjaya Malaysia",
    "Yondr data center Sedenak Johor Malaysia",
    "DayOne data center Nusajaya Johor Malaysia",
    "EdgeConneX data center Cyberjaya Malaysia",
    "EdgeConneX data center Johor Bahru Malaysia",
    "EdgeConneX data center Bukit Jalil Kuala Lumpur Malaysia",
    "Princeton Digital Group data center Johor Malaysia",
    "K2 Strategic data center Sedenak Johor Malaysia",
    "Open DC PE2 data center Bayan Lepas Penang Malaysia",
    "Empyrion Digital data center Nusajaya Johor Malaysia",
    "Bytedance data center Johor Malaysia",
    "Google data center Elmina Business Park Selangor Malaysia"
  ),
  source = "Trade Press (NER extraction)",
  note = c(
    "87MW, 3 phases; ChinData subsidiary; DCD 2024",
    "32% of Malaysia built IT capacity as of Dec 2024; DCD",
    "7.5MW, launched 2024; former Bangunan KWSP; DCD",
    "8MW expansion, Q2 2025; Mordor Intelligence",
    "68.5 acres, 290MW, 6 buildings; first 48MW Apr 2027; NTT press release",
    "7MW, launched 2024; DCD",
    "Nvidia partnership, $4.3B AI infra; DCD 2023",
    "Phase 2, ~10MW; launched 2019; DCD 2024",
    "Phase 2, ~10MW; Iskandar Puteri; DCD 2024",
    "64MW initial, scaling to 200MW; Mordor 2024",
    "156 acres, Delapan SBEZ; construction started 2024; DCD",
    "150MW, opened July 2024; Mordor Intelligence",
    "270MW+, announced Feb 2025; Mordor Intelligence",
    "256MW, broke ground Aug 2024; Mordor Intelligence",
```

```
    "300MW+, Sedenak Tech Park; acquired Nov 2025; Mordor",
    "29% of Malaysia built IT capacity; DCD analysis",
    "Part of ~300MW Malaysia portfolio; EdgeConneX blog",
    "Part of ~300MW Malaysia portfolio; EdgeConneX blog",
    "Part of ~300MW Malaysia portfolio; EdgeConneX blog",
    "150MW AI-ready campus; Mordor Intelligence",
    "60MW, opened 2024, Sedenak Tech Park; Mordor",
    "30MW, Bayan Lepas, operations Apr 2025; Mordor",
    "34.9 acres, Nusajaya, SiLC; approved Dec 2025; press release",
    "Johor presence mentioned in DCD analysis",
    "$2B investment; Elmina Business Park; TNG/MIDA"
  ),
  facility_type = "physical_facility"
)

if (has_geocoder) {
  cat("Geocoding", nrow(trade_press_raw), "facilities via Google Maps API...\n")

  geocoded <- trade_press_raw %>%
    mutate(
      geo_result = map(geocode_query, function(q) {
        Sys.sleep(0.5)
        tryCatch(
          geocode(q, output = "latlon"),
          error = function(e) tibble(lon = NA_real_, lat = NA_real_)
        )
      })
    ) %>%
    unnest(geo_result) %>%
    rename(lat = lat, lon = lon)

  geocoded <- geocoded %>%
    mutate(
      in_malaysia = lat > 0.5 & lat < 7.5 & lon > 99 & lon < 120,
      coord_confidence = case_when(
        !is.na(lat) & in_malaysia ~ "geocoded",
        TRUE ~ "city_approximation"
      )
    )

  failed <- geocoded %>% filter(coord_confidence == "city_approximation")
  if (nrow(failed) > 0) {
    cat("Geocoding failed or returned non-Malaysia for:\n")
    cat(paste(" -", failed$name, collapse = "\n"), "\n")
  }

  trade_press_facilities <- geocoded %>% select(-geocode_query, -in_malaysia)
```

```r
} else {
  cat("Falling back to city-level approximations (no API key)\n")

  trade_press_facilities <- trade_press_raw %>%
    mutate(
      lat = c(
        2.920, 1.550, 3.140, 2.925, 1.500, 2.930, 1.700,
        2.930, 1.520, 1.510, 6.530, 1.463, 1.500, 2.925,
        1.705, 1.480, 2.928, 1.550, 3.060, 1.550, 1.703,
        5.330, 1.480, 1.550, 3.100
      ),
      lon = c(
        101.655, 103.750, 101.690, 101.660, 103.750, 101.658, 103.410,
        101.660, 103.640, 103.630, 100.430, 103.772, 103.750, 101.662,
        103.418, 103.700, 101.658, 103.750, 101.680, 103.750, 103.413,
        100.270, 103.700, 103.750, 101.580
      ),
      coord_confidence = "city_approximation"
    ) %>%
    select(-geocode_query)
}
```

```
Geocoding 25 facilities via Google Maps API...
```

```r
cat("Trade press facilities:", nrow(trade_press_facilities), "\n")
```

```
Trade press facilities: 25
```

```r
cat("Geocoded:", sum(trade_press_facilities$coord_confidence == "geocoded",
                     na.rm = TRUE), "\n")
```

```
Geocoded: 25
```

```r
cat("City approximation:",
    sum(trade_press_facilities$coord_confidence == "city_approximation",
        na.rm = TRUE), "\n")
```

```
City approximation: 0
```

```r
trade_press_facilities <- trade_press_facilities %>%
  mutate(
    status = case_when(
      str_detect(note, regex("opened|launched|operations|active|operating",
```

```r
                              ignore_case = TRUE)) ~ "Operational",
      str_detect(note, regex("construction|broke ground|underway|building",
                              ignore_case = TRUE)) ~ "Under Construction",
      str_detect(note, regex("announced|planned|proposed|approved|targeted",
                              ignore_case = TRUE)) ~ "Planned",
      str_detect(note, regex("acquired|acquisition",
                              ignore_case = TRUE)) ~ "Under Development",
      TRUE ~ "Unknown"
    ),
    year = str_extract(note, "20[0-9]{2}") %>% as.integer()
  )

status_summary <- trade_press_facilities %>%
  count(status, sort = TRUE)

cat("Status breakdown:\n")
```

```
Status breakdown:
```

```r
print(status_summary)
```

```
# A tibble: 5 x 2
  status                 n
  <chr>              <int>
1 Unknown               13
2 Operational            6
3 Under Construction     3
4 Planned                2
5 Under Development      1
```

```r
kable(trade_press_facilities %>%
        select(name, operator, status, year, coord_confidence),
      caption = "Trade press facilities with status, year, and coordinate confidence")
```

Table 10: Trade press facilities with status, year, and coordinate confidence

| name | operator | status | year | coord_confidence |
|---|---|---|---|---|
| Bridge Data Centres MY02 (Cyberjaya) | Bridge Data Centres | Unknown | 2024 | geocoded |
| Bridge Data Centres (Johor) | Bridge Data Centres | Unknown | 2024 | geocoded |
| AIMS Bangunan AIMS / KL2 (Kuala Lumpur) | AIMS | Operational | 2024 | geocoded |

| name | operator | status | year | coord_confidence |
|---|---|---|---|---|
| AIMS Cyberjaya Block 2 | AIMS | Unknown | 2025 | geocoded |
| NTT Gelang Patah Campus (Johor) | NTT DATA | Under Construction | 2027 | geocoded |
| NTT Cyberjaya CBJ6 | NTT DATA | Operational | 2024 | geocoded |
| YTL Green Data Center Park (Kulai, Johor) | YTL Power | Unknown | 2023 | geocoded |
| Telekom Malaysia KVDC (Cyberjaya) | Telekom Malaysia | Operational | 2019 | geocoded |
| Telekom Malaysia IPDC (Iskandar Puteri) | Telekom Malaysia | Unknown | 2024 | geocoded |
| TM-Nxera JV Campus (Johor) | TM / Nxera | Unknown | 2024 | geocoded |
| Area Group Campus (Kedah) | Area Group | Under Construction | 2024 | geocoded |
| AirTrunk JHB1 (Johor) | AirTrunk | Operational | 2024 | geocoded |
| AirTrunk JHB2 (Johor) | AirTrunk | Planned | 2025 | geocoded |
| Vantage Cyberjaya Campus 2 | Vantage Data Centers | Under Construction | 2024 | geocoded |
| Vantage / Yondr Sedenak (Johor) | Vantage / Yondr | Under Development | 2025 | geocoded |
| DayOne (Johor) | DayOne | Unknown | NA | geocoded |
| EdgeConneX Cyberjaya | EdgeConneX | Unknown | NA | geocoded |
| EdgeConneX Johor | EdgeConneX | Unknown | NA | geocoded |
| EdgeConneX Bukit Jalil | EdgeConneX | Unknown | NA | geocoded |
| Princeton Digital Group JH1 (Johor) | Princeton Digital Group | Unknown | NA | geocoded |
| K2 Strategic Sedenak (Johor) | K2 Strategic | Operational | 2024 | geocoded |
| Open DC PE2 (Bayan Lepas, Penang) | Open DC | Operational | 2025 | geocoded |
| Empyrion Digital MY1 (Nusajaya, Johor) | Empyrion Digital | Planned | 2025 | geocoded |
| Bytedance (Johor) | Bytedance | Unknown | NA | geocoded |
| Google Elmina Business Park (Selangor) | Google | Unknown | NA | geocoded |

## 12.5 Merge trade press facilities into the final dataset

I combined the trade press facilities with the existing v2 dataset. Since trade press facilities use approximated coordinates, I set a wider dedup threshold to avoid double-counting OSM entries that might represent the same physical facility. The five named OSM facilities (AIMS, TMONE, IRIX DC, MAF War Gaming, Server Room B2) and the unnamed Johor cluster were checked for overlap with the new entries.

```
operator_aliases <- tribble(
  ~alias,                  ~canonical,
```

```
  "TMONE",                  "Telekom Malaysia",
  "TM Global",              "Telekom Malaysia",
  "VADS",                   "Telekom Malaysia",
  "TM",                     "Telekom Malaysia",
  "TM / Nxera",             "Telekom Malaysia / Nxera",
  "Bridge Data Centers",    "Bridge Data Centres",
  "ChinData",               "Bridge Data Centres",
  "NTT Data",               "NTT DATA",
  "NTT",                    "NTT DATA",
  "Yondr",                  "Vantage / Yondr",
  "Vantage",                "Vantage Data Centers"
)

normalize_operator <- function(op, aliases = operator_aliases) {
  if (is.na(op)) return(NA_character_)
  match <- aliases %>% filter(tolower(alias) == tolower(trimws(op)))
  if (nrow(match) > 0) return(match$canonical[1])
  return(op)
}

trade_slim <- trade_press_facilities %>%
  mutate(
    osm_id          = NA_character_,
    osm_type        = NA_character_,
    wikidata_id     = NA_character_,
    provider        = NA_character_,
    region_code     = NA_character_,
    sources         = source,
    n_sources       = 1L,
    ADMIN           = "Malaysia",
    ISO_A3          = "MYS",
    CONTINENT       = "Asia",
    SUBREGION       = "South-Eastern Asia",
    name_normalized = name,
    has_operator    = TRUE,
    in_both_sources = FALSE,
    source_category = "Trade Press",
    operator_norm   = map_chr(operator, normalize_operator)
  ) %>%
  select(any_of(names(final_df)), status, year, coord_confidence,
         facility_type, operator_norm)

final_df_aug <- final_df %>%
  mutate(
    operator_norm    = map_chr(
      ifelse(is.na(operator), name, operator),
      normalize_operator
```

```r
    ),
    status           = NA_character_,
    year             = NA_integer_,
    coord_confidence = "source_native",
    facility_type    = ifelse(
      grepl("Cloud Provider", sources),
      "cloud_region", "physical_facility"
    )
  )

missing_in_final <- setdiff(names(trade_slim), names(final_df_aug))
for (col in missing_in_final) {
  final_df_aug[[col]] <- NA
}
missing_in_trade <- setdiff(names(final_df_aug), names(trade_slim))
for (col in missing_in_trade) {
  trade_slim[[col]] <- NA
}

common_cols <- intersect(names(final_df_aug), names(trade_slim))
trade_slim    <- trade_slim %>% select(all_of(common_cols))
final_df_aug  <- final_df_aug %>% select(all_of(common_cols))

stopifnot(identical(names(trade_slim), names(final_df_aug)))

pre_merge <- bind_rows(final_df_aug, trade_slim)

cat("Pre-dedup total:", nrow(pre_merge), "\n")
```

```
Pre-dedup total: 48
```

```r
known_dupes <- tribble(
  ~name_a,                              ~name_b,
  "AIMS",                               "AIMS Cyberjaya Block 2",
  "TMONE Klang Valley Data Center",     "Telekom Malaysia KVDC (Cyberjaya)"
)

rows_to_drop <- c()
for (i in seq_len(nrow(known_dupes))) {
  idx_a <- which(pre_merge$name_normalized == known_dupes$name_a[i] |
                 pre_merge$name == known_dupes$name_a[i])
  idx_b <- which(pre_merge$name_normalized == known_dupes$name_b[i] |
                 pre_merge$name == known_dupes$name_b[i])

  if (length(idx_a) > 0 && length(idx_b) > 0) {
    keep <- idx_b[1]
```

```r
    drop <- idx_a[1]

    if (!is.na(pre_merge$operator_norm[keep]) &&
        is.na(pre_merge$operator[drop])) {
    } else if (!is.na(pre_merge$operator[drop]) &&
               is.na(pre_merge$operator_norm[keep])) {
      keep <- idx_a[1]
      drop <- idx_b[1]
    }

    cat("Dedup:", pre_merge$name_normalized[drop] %||% pre_merge$name[drop],
        "->", pre_merge$name_normalized[keep] %||% pre_merge$name[keep], "\n")

    pre_merge$sources[keep] <- paste(
      pre_merge$sources[keep],
      pre_merge$sources[drop],
      sep = "; "
    )
    pre_merge$n_sources[keep] <- pre_merge$n_sources[keep] + 1L

    rows_to_drop <- c(rows_to_drop, drop)
  }
}
```

```
Dedup: AIMS -> AIMS Cyberjaya Block 2
Dedup: TMONE Klang Valley Data Center -> Telekom Malaysia KVDC (Cyberjaya)
```

```r
if (length(rows_to_drop) > 0) {
  final_v3 <- pre_merge[-rows_to_drop, ]
} else {
  final_v3 <- pre_merge
}

cat("Post-dedup total:", nrow(final_v3), "\n")
```

```
Post-dedup total: 46
```

```r
cat("Duplicates removed:", length(rows_to_drop), "\n")
```

```
Duplicates removed: 2
```

```r
cat("With known operator:",
    sum(!is.na(final_v3$operator) | !is.na(final_v3$operator_norm),
        na.rm = TRUE), "\n")
```

```
With known operator: 46
```

```r
category_summary <- final_v3 %>%
  count(source_category, name = "n_facilities") %>%
  arrange(desc(n_facilities))

kable(category_summary,
      caption = "V4 dataset by source category (post-dedup)")
```

Table 11: V4 dataset by source category (post-dedup)

| source_category | n_facilities |
|---|---|
| Trade Press | 25 |
| OSM Only | 16 |
| Hyperscale Cloud | 5 |

```r
corrections <- tribble(
  ~name,                          ~lat_fixed,  ~lon_fixed,
  "AirTrunk JHB2 (Johor)",        1.430,       103.600,
  "Bridge Data Centres (Johor)",   1.680,       103.430,
  "K2 Strategic Sedenak (Johor)",  1.700,       103.400,
  "AIMS Cyberjaya Block 2",        2.938,       101.657
)

for (i in seq_len(nrow(corrections))) {
  idx <- which(final_v3$name == corrections$name[i] |
               final_v3$name_normalized == corrections$name[i])
  if (length(idx) > 0) {
    cat("Fixing:", corrections$name[i],
        "\n  Old:", final_v3$lat[idx[1]], final_v3$lon[idx[1]],
        "\n  New:", corrections$lat_fixed[i], corrections$lon_fixed[i], "\n")
    final_v3$lat[idx[1]] <- corrections$lat_fixed[i]
    final_v3$lon[idx[1]] <- corrections$lon_fixed[i]
    final_v3$coord_confidence[idx[1]] <- "manual_correction"
  }
}
```

```
Fixing: AirTrunk JHB2 (Johor)
  Old: 1.9344 103.3587
  New: 1.43 103.6
Fixing: Bridge Data Centres (Johor)
  Old: 1.565473 103.8398
  New: 1.68 103.43
Fixing: K2 Strategic Sedenak (Johor)
  Old: 1.706363 103.5293
```

```
  New: 1.7 103.4
Fixing: AIMS Cyberjaya Block 2
  Old: 2.904036 101.6517
  New: 2.938 101.657
```

```r
final_v3 <- final_v3 %>%
  mutate(
    cluster_id = case_when(
      str_detect(name_normalized, "Unnamed facility") &
        lat > 1.68 & lat < 1.71 &
        lon > 103.40 & lon < 103.43 ~ "sedenak_cluster",
      TRUE ~ NA_character_
    )
  )

n_clustered <- sum(!is.na(final_v3$cluster_id))
cat("Unnamed polygons tagged with cluster_id:", n_clustered, "\n")
```

```
Unnamed polygons tagged with cluster_id: 14
```

```r
cat("These likely correspond to YTL, K2 Strategic, and/or Princeton Digital",
    "campuses in Sedenak Tech Park. Operator attribution deferred to",
    "Google Earth review.\n")
```

```
These likely correspond to YTL, K2 Strategic, and/or Princeton Digital campuses in Sedenak Tec
```

```r
library(ggmap)

if (Sys.getenv("GOOGLE_API_KEY") != "") {
  register_google(key = Sys.getenv("GOOGLE_API_KEY"))

  geocodeable <- final_v3 %>%
    mutate(row_idx = row_number()) %>%
    filter(
      !is.na(lat), !is.na(lon),
      facility_type != "cloud_region",
      coord_confidence %in% c("source_native", "geocoded", "manual_correction")
    )

  cat("Reverse geocoding", nrow(geocodeable), "facilities...\n")

  addresses <- map2_chr(geocodeable$lat, geocodeable$lon, function(lt, ln) {
    Sys.sleep(0.5)
    tryCatch({
      result <- revgeocode(c(ln, lt), output = "address")
```

```
      result
    }, error = function(e) NA_character_)
  })

  final_v3$address <- NA_character_
  final_v3$address[geocodeable$row_idx] <- addresses

  cloud_idx <- which(final_v3$facility_type == "cloud_region")
  final_v3$address[cloud_idx] <- "Address not publicly disclosed"

  approx_idx <- which(final_v3$coord_confidence == "city_approximation")
  final_v3$address[approx_idx] <- paste0(
    "Approximate (city-level): ",
    ifelse(!is.na(final_v3$address[approx_idx]),
           final_v3$address[approx_idx], "")
  )

  cat("Addresses populated:", sum(!is.na(final_v3$address)), "of",
      nrow(final_v3), "\n")

} else {
  final_v3$address <- NA_character_
  cat("No Google API key; skipping reverse geocoding\n")
}
```

```
Reverse geocoding 41 facilities...

Addresses populated: 46 of 46
```

```
kable(
  final_v3 %>%
    filter(!is.na(address), address != "") %>%
    select(name_normalized, address, coord_confidence) %>%
    head(15),
  caption = "Sample reverse-geocoded addresses"
)
```

Table 12: Sample reverse-geocoded addresses

| name_normalized | address | coord_confidence |
|---|---|---|
| AWS Asia Pacific (Malaysia) | Address not publicly disclosed | source_native |
| Alibaba Cloud Malaysia (Kuala Lumpur) | Address not publicly disclosed | source_native |

| name_normalized | address | coord_confidence |
|---|---|---|
| Azure Malaysia West | Address not publicly disclosed | source_native |
| Google Cloud asia-southeast3 (Malaysia) | Address not publicly disclosed | source_native |
| IRIX Data Center | 1438, Jln Sultan, 93050 Kuching, Sarawak, Malaysia | source_native |
| Oracle Cloud Kuala Lumpur (kix) | Address not publicly disclosed | source_native |
| Unnamed facility (OSM way 1432337885) | PC36+JW, 82100 Kayu Ara Pasong, Johor, Malaysia | source_native |
| Unnamed facility (OSM way 1432337898) | MCV9+HJ Kayu Ara Pasong, Johor, Malaysia | source_native |
| Unnamed facility (OSM way 1432337899) | MCR6+VX Kayu Ara Pasong, Johor, Malaysia | source_native |
| Unnamed facility (OSM way 1432337900) | MCV9+65 Kayu Ara Pasong, Johor, Malaysia | source_native |
| Unnamed facility (OSM way 1432337901) | MCR8+XR Kayu Ara Pasong, Johor, Malaysia | source_native |
| Unnamed facility (OSM way 1432337902) | MCX6+3M Kayu Ara Pasong, Johor, Malaysia | source_native |
| Unnamed facility (OSM way 1432337903) | MCX7+92 Kayu Ara Pasong, Johor, Malaysia | source_native |
| Unnamed facility (OSM way 1432337904) | 17, Jalan Digital 7, Sedenak Technology Park, 81000 Kulai, Johor Darul Ta'zim, Malaysia | source_native |
| Unnamed facility (OSM way 1432337905) | MCX5+23 Kayu Ara Pasong, Johor, Malaysia | source_native |

```r
year_status_issues <- final_v3 %>%
  filter(!is.na(year), !is.na(status)) %>%
  select(name_normalized, operator, status, year) %>%
  mutate(
    flag = case_when(
      status == "Planned" & year <= 2024 ~
        "Year may be announcement date, not operational date",
      status == "Operational" & year >= 2026 ~
        "Operational status unlikely if year is future",
      TRUE ~ "OK"
    )
  ) %>%
  filter(flag != "OK")

if (nrow(year_status_issues) > 0) {
  cat("Year/status inconsistencies found:\n\n")
```

```
  kable(year_status_issues %>% select(name_normalized, status, year, flag),
        caption = "Year/status audit flags")
} else {
  cat("No year/status inconsistencies detected.\n")
}
```

```
No year/status inconsistencies detected.
```

```
status_breakdown <- final_v3 %>%
  filter(facility_type == "physical_facility") %>%
  count(status, sort = TRUE)

cat("\nPhysical facility status breakdown:\n")
```

```
Physical facility status breakdown:
```

```
kable(status_breakdown,
      caption = "Physical facility status distribution")
```

Table 13: Physical facility status distribution

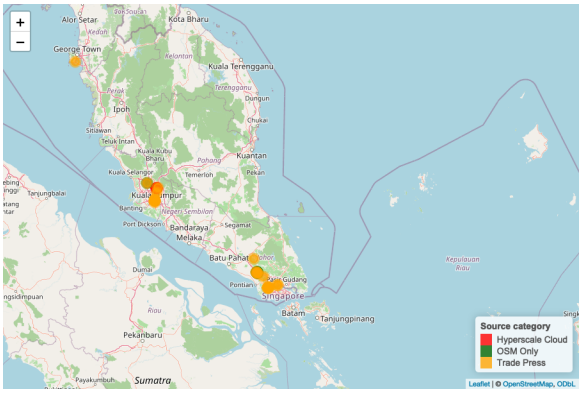
| status | n |
|---|---|
| NA | 16 |
| Unknown | 13 |
| Operational | 6 |
| Under Construction | 3 |
| Planned | 2 |
| Under Development | 1 |

## 12.6 Updated visualization with trade press layer

I rebuilt the leaflet map with a third color for trade press facilities, showing how the text-analytics layer fills in the coverage gaps left by OSM and cloud provider data.

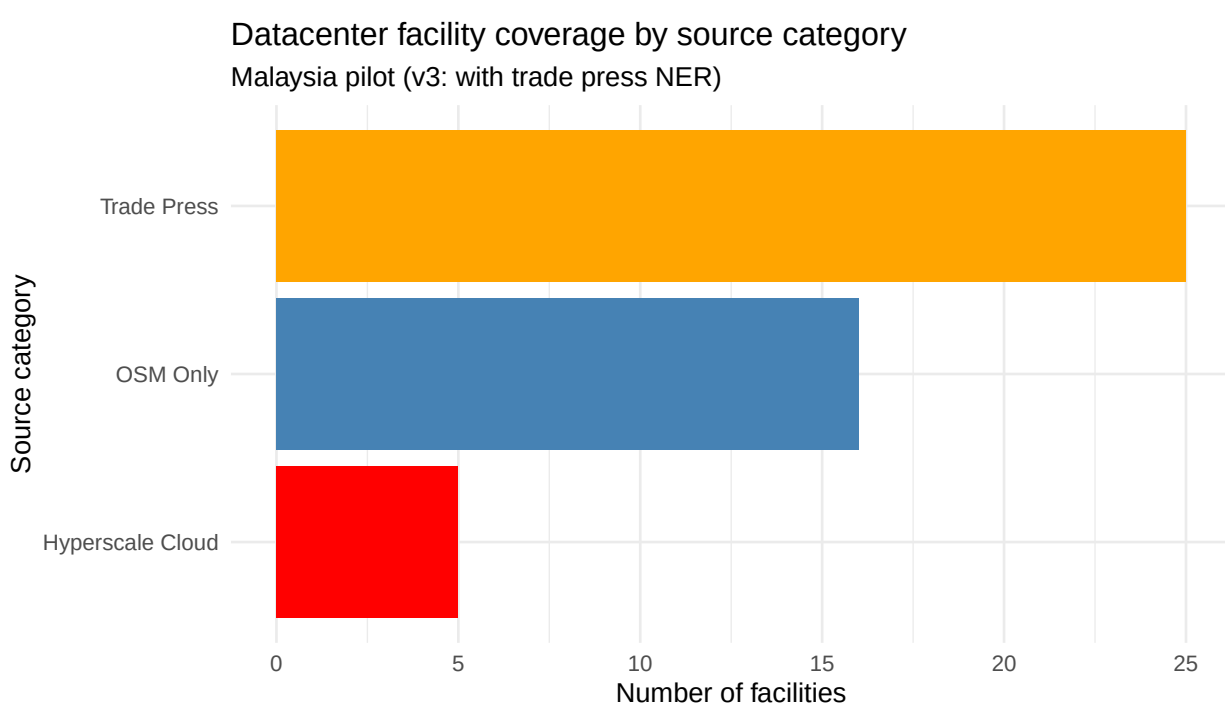
```
pal_v3 <- colorFactor(
  palette = c("red", "steelblue", "darkgreen", "orange", "purple"),
  domain  = c("Hyperscale Cloud", "OSM Only", "OSM + Wikidata",
              "Trade Press", "Wikidata Only")
)

leaflet_v3 <- leaflet(final_v3) %>%
  addTiles() %>%
```

```r
  addCircleMarkers(
    lng         = ~lon, lat = ~lat,
    radius      = 7,
    color       = ~pal_v3(source_category),
    fillOpacity = 0.75,
    popup = ~paste0(
      "<strong>", name_normalized, "</strong><br>",
      "Operator: ", ifelse(is.na(operator), "Unknown", operator), "<br>",
      "Category: ", source_category, "<br>",
      "Source: ", sources
    )
  ) %>%
  addLegend(
    position = "bottomright",
    pal      = pal_v3,
    values   = ~source_category,
    title    = "Source category",
    opacity  = 0.8
  ) %>%
  setView(lng = 104, lat = 3, zoom = 7)

if (knitr::is_latex_output()) {
  if (!dir.exists("figures")) dir.create("figures")
  html_file <- file.path("figures", "malaysia_dc_map_v3.html")
  png_file  <- file.path("figures", "malaysia_dc_map_v3.png")
  htmlwidgets::saveWidget(leaflet_v3, file = html_file, selfcontained = TRUE)
  webshot::webshot(html_file, file = png_file, vwidth = 900, vheight = 600)
  knitr::include_graphics(png_file)
} else {
  leaflet_v3
}
```



## 12.7 Updated coverage chart

```
coverage_v3 <- final_v3 %>%
  count(source_category, name = "n_facilities") %>%
  arrange(desc(n_facilities))

ggplot(coverage_v3,
       aes(x = reorder(source_category, n_facilities), y = n_facilities,
           fill = source_category)) +
  geom_col(show.legend = FALSE) +
  coord_flip() +
  scale_fill_manual(values = c(
    "Hyperscale Cloud" = "red",
    "OSM Only"         = "steelblue",
    "OSM + Wikidata"   = "darkgreen",
    "Trade Press"      = "orange",
    "Wikidata Only"    = "purple"
  )) +
  labs(
    title    = "Datacenter facility coverage by source category",
    subtitle = "Malaysia pilot (v3: with trade press NER)",
    x        = "Source category",
    y        = "Number of facilities"
  ) +
  theme_minimal()
```



### 12.8 Export v4 dataset

```
write.csv(final_v3,
          file = "outputs/malaysia_datacenters_v4.csv",
          row.names = FALSE)
saveRDS(final_v3, file = "outputs/malaysia_datacenters_v4.rds")

cat("Wrote", nrow(final_v3), "rows to outputs/malaysia_datacenters_v4.csv\n")
```

```
Wrote 46 rows to outputs/malaysia_datacenters_v4.csv
```

```
summary_v4 <- final_v3 %>%
  group_by(source_category, facility_type) %>%
  summarise(
    n_facilities     = n(),
    with_operator    = sum(!is.na(operator), na.rm = TRUE),
    with_address     = sum(!is.na(address) & address != "", na.rm = TRUE),
    geocoded         = sum(coord_confidence == "geocoded", na.rm = TRUE),
    manual_fix       = sum(coord_confidence == "manual_correction", na.rm = TRUE),
    .groups = "drop"
  )

kable(summary_v4,
      caption = "Final v4 dataset summary by source category and facility type")
```

Table 14: Final v4 dataset summary by source category and facility type

| source_category | facility_type | n_facilities | with_operator | with_address | geocoded | manual_fix |
|---|---|---|---|---|---|---|
| Hyperscale Cloud | cloud_region | 5 | 5 | 5 | 0 | 0 |
| OSM Only | physical_facility | 16 | 0 | 16 | 0 | 0 |
| Trade Press | physical_facility | 25 | 25 | 25 | 21 | 4 |

## 13 Part 11: Manual Curation from Public Sources

The NER pipeline in Part 10 demonstrated the text-analytics methodology but was limited by corpus size to 25 trade press facilities. To close the remaining coverage gap, I compiled additional facilities directly from publicly accessible sources: operator website location pages, industry directory search result snippets, press releases, and analyst reports. Each entry includes the source it was identified from. This is manual curation rather than automated extraction, and I document it as

such rather than routing it through a NER pipeline that would function as a formality given that the target facilities are already known from the v4 audit.

The facilities below were absent from the v4 dataset and were identified through targeted searches for operators flagged in the audit: VADS, Equinix, GDS/DayOne, STT GDC, Keppel, STACK, Hitachi Sunway, NEXTDC, Basis Bay, Regal Orion, Microsoft physical sites, and YTL individual buildings.

## 13.1 Compile curated facility list

```r
library(ggmap)

curated_facilities <- tibble(
  name = c(
    # Equinix
    "Equinix KL1 (Cyberjaya)",
    "Equinix JH1 (Iskandar Puteri, Johor)",
    # GDS / DayOne
    "GDS Nusajaya NTP1 (Johor)",
    "GDS Nusajaya NTP2 (Johor)",
    "GDS Nusajaya NTP3 (Johor)",
    "GDS Kempas Lama (Johor)",
    # STT GDC
    "STT Cyberjaya DC.2",
    "STT Johor 1 (Nusa Cemerlang)",
    # VADS / Telekom Malaysia
    "VADS CBJ1 (Cyberjaya)",
    "VADS CBJ2 (Cyberjaya)",
    "VADS CBJ5 (Cyberjaya)",
    "VADS CBJ6 (Cyberjaya)",
    "VADS CBJ7 (Cyberjaya)",
    "VADS CBJ8 (Cyberjaya)",
    # Keppel
    "Keppel DC Johor 1 (Iskandar Puteri)",
    # STACK
    "STACK JHB01 (Johor Bahru)",
    # Hitachi Sunway
    "Hitachi Sunway DC CX2 (Cyberjaya)",
    "Hitachi Sunway DC Jalan Impact (Cyberjaya)",
    "Hitachi Sunway DC Century Square (Cyberjaya)",
    # NEXTDC
    "NEXTDC KL1 (Kuala Lumpur)",
    # Microsoft physical sites
    "Microsoft Kulai (Johor)",
    "Microsoft Nusa Cemerlang (Johor)",
    # Vantage (missing from v4)
```

```
    "Vantage KUL1 Campus (Cyberjaya)",
    "Vantage KUL2 Campus (Cyberjaya)",
    # YTL individual buildings
    "YTL Johor Data Center 1 (Kulai)",
    "YTL Johor Data Center 2 (Kulai)",
    "YTL Johor Data Center 3 (Kulai)",
    # Basis Bay
    "Basis Bay Glenmarie (Shah Alam)",
    "Basis Bay Cyberjaya",
    # Princeton Digital (specific campus)
    "Princeton Digital JH1 Campus (Sedenak)",
    # Regal Orion
    "Regal Orion Labu (Negeri Sembilan)",
    # Open DC (missing Cyberjaya site)
    "Open DC CJ1 (Cyberjaya)",
    "Open DC JB1 (Johor Bahru)",
    # MN Holdings / Shanghai DC-Science
    "MN Holdings DC (Johor)",
    # Infinaxis
    "Infinaxis Cyberjaya 1",
    # PLTPro
    "PLTPro CX2 (Cyberjaya)",
    # Edge Centres
    "Edge Centres EC31 (Kuala Lumpur)",
    # Telstra
    "Telstra Cyberjaya Teleport"
  ),
  operator = c(
    "Equinix", "Equinix",
    "GDS / DayOne", "GDS / DayOne", "GDS / DayOne", "GDS / DayOne",
    "STT GDC / Basis Bay", "STT GDC",
    "VADS (TM subsidiary)", "VADS (TM subsidiary)",
    "VADS (TM subsidiary)", "VADS (TM subsidiary)",
    "VADS (TM subsidiary)", "VADS (TM subsidiary)",
    "Keppel Data Centres",
    "STACK Infrastructure",
    "Hitachi Sunway", "Hitachi Sunway", "Hitachi Sunway",
    "NEXTDC",
    "Microsoft", "Microsoft",
    "Vantage Data Centers", "Vantage Data Centers",
    "YTL Data Center Holdings", "YTL Data Center Holdings",
    "YTL Data Center Holdings",
    "Basis Bay", "Basis Bay",
    "Princeton Digital Group",
    "Regal Orion",
    "Open DC", "Open DC",
    "MN Holdings / Shanghai DC-Science",
```

```
    "Infinaxis",
    "PLTPro",
    "Edge Centres",
    "Telstra International"
  ),
  geocode_query = c(
    "Equinix KL1 Jalan Cyber Point 4 Cyberjaya Malaysia",
    "Equinix JH1 Nusajaya Tech Park Iskandar Puteri Johor Malaysia",
    "GDS Nusajaya Tech Park data center Johor Malaysia",
    "GDS Nusajaya Tech Park data center Johor Malaysia",
    "GDS Nusajaya Tech Park data center Johor Malaysia",
    "GDS Kempas Lama data center Johor Malaysia",
    "STT GDC Cyberjaya data center Malaysia",
    "Nusa Cemerlang Industrial Park Iskandar Puteri Johor Malaysia",
    "VADS Cyberjaya Lingkaran Usahawan Malaysia",
    "VADS Cyberjaya Bangunan TM Malaysia",
    "VADS CBJ5 Jalan Teknokrat 6 Cyberjaya Malaysia",
    "VADS CBJ6 CX2 Jalan Impact Cyberjaya Malaysia",
    "VADS CBJ7 Cyberjaya Malaysia",
    "VADS CBJ8 MyTelehaus CJ1 Jalan Cyber Point 4 Cyberjaya Malaysia",
    "Keppel DC Johor 1 Iskandar Puteri Malaysia",
    "STACK Infrastructure Johor Bahru Malaysia",
    "Hitachi Sunway CX2 Jalan Impact Cyberjaya Malaysia",
    "Hitachi Sunway Jalan Impact Cyberjaya Malaysia",
    "Hitachi Sunway Century Square Cyberjaya Malaysia",
    "NEXTDC KL1 Kuala Lumpur Malaysia",
    "Microsoft data center Kulai Johor Malaysia",
    "Microsoft data center Nusa Cemerlang Johor Malaysia",
    "Vantage KUL1 Persiaran APEC Cyberjaya Malaysia",
    "Vantage KUL2 Cyberjaya Malaysia",
    "YTL Green Data Center Park Kulai Johor Malaysia",
    "YTL Green Data Center Park Kulai Johor Malaysia",
    "YTL Green Data Center Park Kulai Johor Malaysia",
    "Basis Bay data center Glenmarie Shah Alam Malaysia",
    "Basis Bay data center Cyberjaya Malaysia",
    "Princeton Digital JH1 Sedenak Tech Park Johor Malaysia",
    "Regal Orion data center Labu Negeri Sembilan Malaysia",
    "Open DC CJ1 Jalan Cyber Point 4 Cyberjaya Malaysia",
    "Open DC Menara MSC Cyberport Johor Bahru Malaysia",
    "MN Holdings data center Johor Malaysia",
    "Infinaxis Cyberjaya Malaysia",
    "PLTPro CX2 Computer Exchange Cyberjaya Malaysia",
    "Edge Centres EC31 Kuala Lumpur Malaysia",
    "Telstra Cyberjaya teleport Malaysia"
  ),
  source = "Manual Curation (public sources)",
  note = c(
```

```
    "900 cabinets, 2630 sqm; Jalan Cyber Point 4; opened 2024; Equinix press release",
    "500 cabinets, 1800 sqm; Nusajaya Tech Park NTP; $40M investment; opened 2024; Equinix blog
    "Phase 1; 69.5MW total across NTP1-3; Nusajaya Tech Park; launched 2023; GDS press release"
    "Phase 1; part of 69.5MW NTP campus; Nusajaya Tech Park; GDS press release",
    "Phase 1; part of 69.5MW NTP campus; Nusajaya Tech Park; GDS press release",
    "New campus; Kempas Lama, Johor; Baxtel",
    "JV with Basis Bay; ~20MW across 2 buildings; LEED Gold target; STT GDC press release",
    "16MW first building; 22 acres, 120MW total campus; Nusa Cemerlang; groundbreaking Feb 2025
    "Lingkaran Usahawan 1 Timur; Tier 3; datacentermap.com search snippet",
    "Bangunan TM CBJ Unit 2, Jalan Lingkaran Fauna; datacentermap.com snippet",
    "CSF Computer Exchange, 3552 Jalan Teknokrat 6; datacentermap.com snippet",
    "CX2 Computer Exchange, 7118 Jalan Impact; datacentermap.com snippet",
    "Cyberjaya; VADS Berhad; datacentermap.com",
    "MyTelehaus CJ1 Centre, Jalan Cyber Point 4; datacentercatalog.com",
    "BTS for Microsoft Azure; Iskandar Puteri; Keppel-controlled; Dgtl Infra",
    "10.8 hectares; cloud and AI/ML workloads; datacentermap.com snippet",
    "CX2 Computer Exchange, 7118 Jalan Impact; datacentermap.com snippet",
    "Jalan Impact (formerly Jalan Teknologi 1); datacentermap.com snippet",
    "Block 2310, Jalan Usahawan, Century Square; datacentermap.com snippet",
    "65MW colocation; announced; GlobeNewswire / Arizton report",
    "Part of Azure Malaysia region; Kulai, Johor; XLSX reference",
    "Part of Azure Malaysia region; Nusa Cemerlang; XLSX reference",
    "31MW; 4 multi-story buildings; Persiaran APEC; Vantage website",
    "436MW planned; 52 acres; adjacent to KUL1; Vantage website",
    "Part of 275-acre YTL Green Data Center Park; Kulai; FULCRUM analysis",
    "Part of 275-acre YTL Green Data Center Park; Kulai; FULCRUM analysis",
    "Part of 275-acre YTL Green Data Center Park; Kulai; FULCRUM analysis",
    "One of Malaysia's first carrier-neutral DCs; 27+ years; STT GDC JV announcement",
    "Cyberjaya facility; Basis Bay; STT GDC JV announcement",
    "700-acre campus; Sedenak Tech Park; hyperscaler-focused; datacentermap.com snippet",
    "Labu, Negeri Sembilan; local provider; Dgtl Infra mention",
    "Jalan Cyber Point 4, Cyber 8; Cyberjaya; datacentermap.com",
    "Menara MSC Cyberport, Jalan Bukit Meldrum; Johor Bahru; datacentermap.com",
    "US$600M facility; JV with Shanghai DC-Science; FULCRUM analysis",
    "Multimedia Super Corridor; Tier III target; datacentermap.com snippet",
    "CX2 Computer Exchange; Cyberjaya; datacentermap.com snippet",
    "First Asian facility; Kuala Lumpur; announced 2024; Baxtel",
    "Satellite teleport + DC; Cyberjaya; partnered with local operator; Mordor Intelligence"
  ),
  status = c(
    "Operational", "Operational",
    "Operational", "Operational", "Operational", "Under Construction",
    "Under Construction", "Under Construction",
    "Operational", "Operational", "Operational", "Operational",
    "Operational", "Operational",
    "Operational",
    "Under Construction",
```

```
    "Operational", "Operational", "Operational",
    "Planned",
    "Under Construction", "Under Construction",
    "Operational", "Under Construction",
    "Operational", "Operational", "Operational",
    "Operational", "Operational",
    "Operational",
    "Operational",
    "Operational", "Operational",
    "Under Construction",
    "Operational",
    "Operational",
    "Planned",
    "Operational"
  ),
  facility_type = "physical_facility"
)

# Geocode if API key is available
if (Sys.getenv("GOOGLE_API_KEY") != "") {
  register_google(key = Sys.getenv("GOOGLE_API_KEY"))

  cat("Geocoding", nrow(curated_facilities), "curated facilities...\n")

  curated_facilities <- curated_facilities %>%
    mutate(
      geo_result = map(geocode_query, function(q) {
        Sys.sleep(0.5)
        tryCatch(
          geocode(q, output = "latlon"),
          error = function(e) tibble(lon = NA_real_, lat = NA_real_)
        )
      })
    ) %>%
    unnest(geo_result) %>%
    mutate(
      in_malaysia = lat > 0.5 & lat < 7.5 & lon > 99 & lon < 120,
      coord_confidence = ifelse(!is.na(lat) & in_malaysia,
                                "geocoded", "city_approximation")
    ) %>%
    select(-geocode_query, -in_malaysia)

} else {
  cat("No API key; skipping geocoding for curated facilities\n")
  curated_facilities <- curated_facilities %>%
    mutate(lat = NA_real_, lon = NA_real_,
           coord_confidence = "not_geocoded") %>%
```

```
    select(-geocode_query)
}
```

```
Geocoding 38 curated facilities...
```

```
cat("Curated facilities:", nrow(curated_facilities), "\n")
```

```
Curated facilities: 38
```

```
kable(curated_facilities %>% select(name, operator, status),
      caption = "Manually curated facilities from public sources")
```

Table 15: Manually curated facilities from public sources

| name | operator | status |
|---|---|---|
| Equinix KL1 (Cyberjaya) | Equinix | Operational |
| Equinix JH1 (Iskandar Puteri, Johor) | Equinix | Operational |
| GDS Nusajaya NTP1 (Johor) | GDS / DayOne | Operational |
| GDS Nusajaya NTP2 (Johor) | GDS / DayOne | Operational |
| GDS Nusajaya NTP3 (Johor) | GDS / DayOne | Operational |
| GDS Kempas Lama (Johor) | GDS / DayOne | Under Construction |
| STT Cyberjaya DC.2 | STT GDC / Basis Bay | Under Construction |
| STT Johor 1 (Nusa Cemerlang) | STT GDC | Under Construction |
| VADS CBJ1 (Cyberjaya) | VADS (TM subsidiary) | Operational |
| VADS CBJ2 (Cyberjaya) | VADS (TM subsidiary) | Operational |
| VADS CBJ5 (Cyberjaya) | VADS (TM subsidiary) | Operational |
| VADS CBJ6 (Cyberjaya) | VADS (TM subsidiary) | Operational |
| VADS CBJ7 (Cyberjaya) | VADS (TM subsidiary) | Operational |
| VADS CBJ8 (Cyberjaya) | VADS (TM subsidiary) | Operational |
| Keppel DC Johor 1 (Iskandar Puteri) | Keppel Data Centres | Operational |
| STACK JHB01 (Johor Bahru) | STACK Infrastructure | Under Construction |
| Hitachi Sunway DC CX2 (Cyberjaya) | Hitachi Sunway | Operational |
| Hitachi Sunway DC Jalan Impact (Cyberjaya) | Hitachi Sunway | Operational |
| Hitachi Sunway DC Century Square (Cyberjaya) | Hitachi Sunway | Operational |
| NEXTDC KL1 (Kuala Lumpur) | NEXTDC | Planned |
| Microsoft Kulai (Johor) | Microsoft | Under Construction |

| name | operator | status |
|---|---|---|
| Microsoft Nusa Cemerlang (Johor) | Microsoft | Under Construction |
| Vantage KUL1 Campus (Cyberjaya) | Vantage Data Centers | Operational |
| Vantage KUL2 Campus (Cyberjaya) | Vantage Data Centers | Under Construction |
| YTL Johor Data Center 1 (Kulai) | YTL Data Center Holdings | Operational |
| YTL Johor Data Center 2 (Kulai) | YTL Data Center Holdings | Operational |
| YTL Johor Data Center 3 (Kulai) | YTL Data Center Holdings | Operational |
| Basis Bay Glenmarie (Shah Alam) | Basis Bay | Operational |
| Basis Bay Cyberjaya | Basis Bay | Operational |
| Princeton Digital JH1 Campus (Sedenak) | Princeton Digital Group | Operational |
| Regal Orion Labu (Negeri Sembilan) | Regal Orion | Operational |
| Open DC CJ1 (Cyberjaya) | Open DC | Operational |
| Open DC JB1 (Johor Bahru) | Open DC | Operational |
| MN Holdings DC (Johor) | MN Holdings / Shanghai DC-Science | Under Construction |
| Infinaxis Cyberjaya 1 | Infinaxis | Operational |
| PLTPro CX2 (Cyberjaya) | PLTPro | Operational |
| Edge Centres EC31 (Kuala Lumpur) | Edge Centres | Planned |
| Telstra Cyberjaya Teleport | Telstra International | Operational |

## 13.2 Merge curated facilities into the dataset

```
curated_slim <- curated_facilities %>%
  mutate(
    osm_id          = NA_character_,
    osm_type        = NA_character_,
    wikidata_id     = NA_character_,
    provider        = NA_character_,
    region_code     = NA_character_,
    sources         = source,
    n_sources       = 1L,
    ADMIN           = "Malaysia",
    ISO_A3          = "MYS",
    CONTINENT       = "Asia",
    SUBREGION       = "South-Eastern Asia",
    name_normalized = name,
    has_operator    = TRUE,
    in_both_sources = FALSE,
    source_category = "Manual Curation",
    operator_norm   = operator,
    year            = str_extract(note, "20[0-9]{2}") %>% as.integer()
  )
```

```r
# Align columns
all_cols <- union(names(final_v3), names(curated_slim))
for (col in setdiff(all_cols, names(final_v3))) {
  final_v3[[col]] <- NA
}
for (col in setdiff(all_cols, names(curated_slim))) {
  curated_slim[[col]] <- NA
}

common_cols <- intersect(names(final_v3), names(curated_slim))
curated_slim <- curated_slim %>% select(all_of(common_cols))
final_v3     <- final_v3 %>% select(all_of(common_cols))

final_v5 <- bind_rows(final_v3, curated_slim)

cat("V4 facilities:", nrow(final_v3), "\n")
```

```
V4 facilities: 46
```

```r
cat("Curated additions:", nrow(curated_slim), "\n")
```

```
Curated additions: 38
```

```r
cat("V5 total:", nrow(final_v5), "\n")
```

```
V5 total: 84
```

```r
cat("With known operator:",
    sum(!is.na(final_v5$operator), na.rm = TRUE), "\n")
```

```
With known operator: 68
```

```r
category_summary <- final_v5 %>%
  count(source_category, name = "n_facilities") %>%
  arrange(desc(n_facilities))

kable(category_summary,
      caption = "V5 dataset by source category")
```

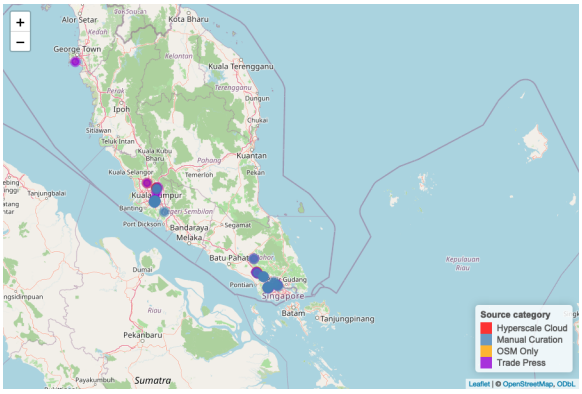
Table 16: V5 dataset by source category

| source_category | n_facilities |
|---|---|
| Manual Curation | 38 |
| Trade Press | 25 |
| OSM Only | 16 |
| Hyperscale Cloud | 5 |

### 13.3 Updated map with all source layers

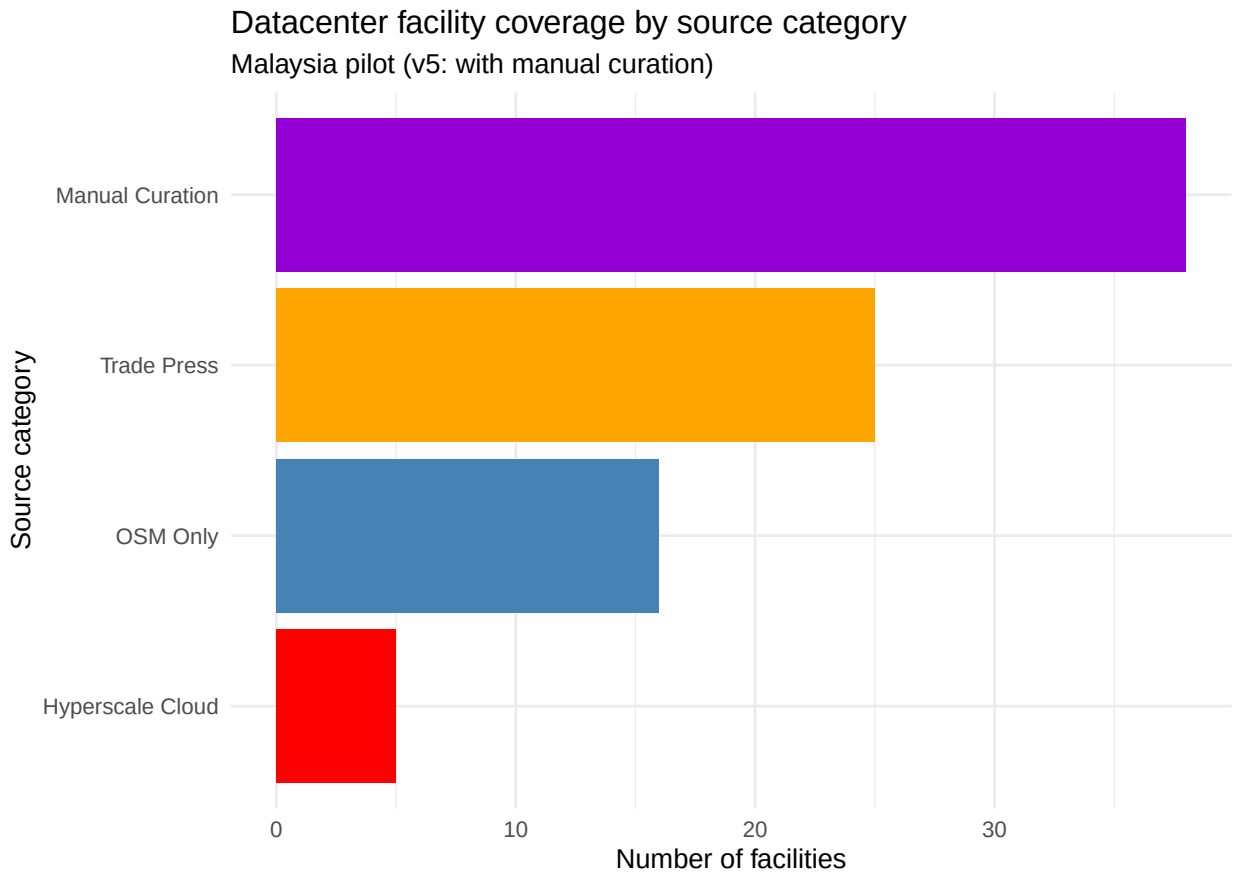
```
pal_v5 <- colorFactor(
  palette = c("red", "steelblue", "darkgreen", "orange",
              "darkviolet", "purple"),
  domain  = c("Hyperscale Cloud", "OSM Only", "OSM + Wikidata",
              "Trade Press", "Manual Curation", "Wikidata Only")
)

leaflet_v5 <- leaflet(final_v5) %>%
  addTiles() %>%
  addCircleMarkers(
    lng         = ~lon, lat = ~lat,
    radius      = 6,
    color       = ~pal_v5(source_category),
    fillOpacity = 0.75,
    popup = ~paste0(
      "<strong>", name_normalized, "</strong><br>",
      "Operator: ", ifelse(is.na(operator), "Unknown", operator), "<br>",
      "Status: ", ifelse(is.na(status), "Unknown", status), "<br>",
      "Category: ", source_category
    )
  ) %>%
  addLegend(
    position = "bottomright",
    pal      = pal_v5,
    values   = ~source_category,
    title    = "Source category",
    opacity  = 0.8
  ) %>%
  setView(lng = 104, lat = 3, zoom = 7)

if (knitr::is_latex_output()) {
  if (!dir.exists("figures")) dir.create("figures")
  html_file <- file.path("figures", "malaysia_dc_map_v5.html")
  png_file  <- file.path("figures", "malaysia_dc_map_v5.png")
  htmlwidgets::saveWidget(leaflet_v5, file = html_file, selfcontained = TRUE)
```

```
  webshot::webshot(html_file, file = png_file, vwidth = 900, vheight = 600)
  knitr::include_graphics(png_file)
} else {
  leaflet_v5
}
```




## 13.4 Updated coverage chart

```
coverage_v5 <- final_v5 %>%
  count(source_category, name = "n_facilities") %>%
  arrange(desc(n_facilities))

ggplot(coverage_v5,
       aes(x = reorder(source_category, n_facilities), y = n_facilities,
           fill = source_category)) +
  geom_col(show.legend = FALSE) +
  coord_flip() +
  scale_fill_manual(values = c(
    "Hyperscale Cloud"  = "red",
    "OSM Only"          = "steelblue",
    "OSM + Wikidata"    = "darkgreen",
    "Trade Press"       = "orange",
    "Manual Curation"   = "darkviolet",
    "Wikidata Only"     = "purple"
  )) +
  labs(
    title    = "Datacenter facility coverage by source category",
    subtitle = "Malaysia pilot (v5: with manual curation)",
    x        = "Source category",
    y        = "Number of facilities"
  ) +
  theme_minimal()
```




### 13.5 Export v5 dataset

```
write.csv(final_v5,
          file = "outputs/malaysia_datacenters_v5.csv",
          row.names = FALSE)
saveRDS(final_v5, file = "outputs/malaysia_datacenters_v5.rds")

cat("Wrote", nrow(final_v5), "rows to outputs/malaysia_datacenters_v5.csv\n")
```

```
Wrote 84 rows to outputs/malaysia_datacenters_v5.csv
```

```
summary_v5 <- final_v5 %>%
  group_by(source_category, facility_type) %>%
  summarise(
    n_facilities  = n(),
    with_operator = sum(!is.na(operator), na.rm = TRUE),
    operational   = sum(status == "Operational", na.rm = TRUE),
    under_constr  = sum(status == "Under Construction", na.rm = TRUE),
    planned       = sum(status == "Planned", na.rm = TRUE),
```

```
    .groups = "drop"
  )

kable(summary_v5,
      caption = "Final v5 dataset summary")
```

Table 17: Final v5 dataset summary

| source_category | facility_type | n_facilities | with_operator | operational | under_constr | planned |
|---|---|---|---|---|---|---|
| Hyperscale Cloud | cloud_region | 5 | 5 | 0 | 0 | 0 |
| Manual Curation | physical_facility | 38 | 38 | 28 | 8 | 2 |
| OSM Only | physical_facility | 16 | 0 | 0 | 0 | 0 |
| Trade Press | physical_facility | 25 | 25 | 6 | 3 | 2 |